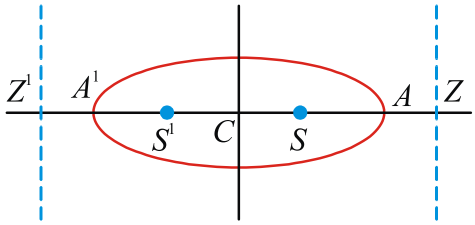
ELLIPSE

SYNOPSIS

Ellipse:

- A conic is said to be an ellipse if its eccentricity is less than 1 $(0<e<1)$
- A second degree non-homogeneous equation $ax^2+2hxy+by^2+2gx+2fy+c=0$ represents an ellipse if $h^2-ab<0$ and $\Delta \neq 0$
- Standard equation of the ellipse is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ .If $a>b$



W.E-1: ***If A, A^1 are the vertices, S, S^1 are the foci and Z, Z^1 are the feet of the directrices of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$ with centre C then CS, CA, CZ are in***

Sol: ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$

$CS=ae, CA=a, CZ=\frac{a}{e}$

$(CA)^2=(CS)(CZ)$ $\therefore$ CS, CA, CZ are in G.P

- **Focal Chord :** A chord passing through either of the foci of the ellipse is called focal chord.
- **Double Ordinate :** A chord passing through a point 'P' on the ellipse perpendicular to the major axis (Principal Axis) of the ellipse is called the double ordinate of the point 'P'.
- **Latusrectum:** A focal chord of an ellipse perpendicular to the major axis (Principal Axis) of the ellipse is called Latusrectum.

W.E-2 : ***The length of latusrectum of the ellipse $9x^2+25y^2-18x-100y-116=0$ is***

Sol: Standard form of given ellipse

$\frac{(x-1)^2}{25}+\frac{(y-2)^2}{9}=1$

$a^2=25 \Rightarrow a=5, \quad b^2=9 \Rightarrow b=3, a>b$

length of latusrectum= $\frac{2b^2}{a}=\frac{2\times 9}{5}=\frac{18}{5}$

Focal Distance :

- The distance of a point 'P' on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ from its focus 'S' is called the focal distance.
- Let 'P' be any point on the ellipse and S,S^1 are foci then SP + S^1P = 2a(a>b). Where SP, S^1P are called focal distances of P.
  i.e sum of the focal distances is equal to length of the major axis.

W.E-3: ***$P\left(\frac{\pi}{6}\right)$ is a point on ellipse $\frac{x^2}{36}+\frac{y^2}{9}=1$ and S and S^1 are the foci of the ellipse. Then $SP+S^1P=$***

Sol: $\frac{x^2}{36}+\frac{y^2}{9}=1$

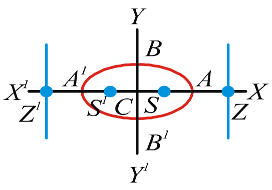
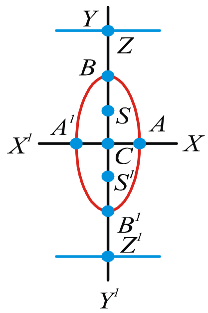
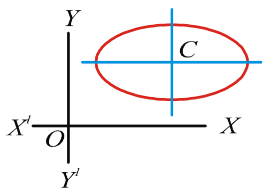
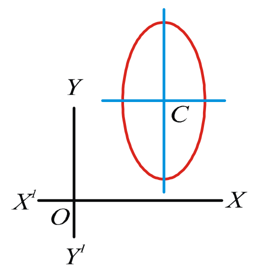
$a^2=36 \Rightarrow a=6, b^2=9 \Rightarrow b=3$

$(a>b), SP+S^1P=2a=2(6)=12$

- If $P(x_1,y_1)$ is a point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,(a>b)$ then the focal distances $SP=|a-ex_1|,\ S^1P=|a+ex_1|$

Note: The focal distances of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\ (a<b)$ are $|b\pm ey_1|$

VARIOUS FORMS OF AN ELLIPSE

	Equation of ellipse	$\frac{x^2}{a^2}+\frac{y^2}{y^2}=1(a>b)$	$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $(a<b)$	$\frac{(x-\alpha)^2}{a^2}+\frac{(y-\beta)^2}{b^2}=1$ $(a>b)$	$\frac{(x-\alpha)^2}{a^2}+\frac{(y-\beta)^2}{b^2}=1$ $(a<b)$
	Figure				
1	Centre (C)	(0,0)	(0,0)	(α, β)	(α, β)
2	Vertices	$(\pm a,0)$	$(0,\pm b)$	$(\alpha\pm a,\beta)$	$(\alpha, \beta\pm b)$
3	Foci (S, S')	$(\pm ae,0)$	$(0,\pm be)$	$(\alpha\pm ae,\beta)$	$(\alpha, \beta\pm be)$
4	Z, Z′ feet of the directrices	$(\pm a/e, 0)$	$(0,\pm b/e)$	$(\alpha\pm a/e, \beta)$	$(\alpha, \beta\pm b/e)$
5	End points of latusrecta	$(\pm ae, \pm b^2/a)$	$(\pm a^2/b, \pm be)$	$(\alpha\pm ae, \beta\pm b^2/a)$	$(\alpha\pm a^2/b, \beta\pm be)$
6	Eqn. of major axis	$y=0$	$x=0$	$y=\beta$	$x=\alpha$
7	Eqn. of minor axis	$x=0$	$y=0$	$x=\alpha$	$y=\beta$
8	Eqn's of latusrecta	$x=\pm ae$	$y=\pm be$	$x=\alpha\pm ae$	$y=\beta\pm be$
9	Eqn's of Directrices	$x=\pm a/e$	$y=\pm b/e$	$x=\alpha\pm a/e$	$y=\beta\pm b/e$
10	Length of major axis	2a	2b	2a	2b
11	Length of minor axis	2b	2a	2b	2a
12	Length of latusrectum	$2b^2/a$	$2a^2/a$	$2b^2/a$	$2a^2/b$
13	Eccentricity (e)	$\sqrt{\frac{a^2-b^2}{a^2}}$	$\sqrt{\frac{b^2-a^2}{b^2}}$	$\sqrt{\frac{a^2-b^2}{a^2}}$	$\sqrt{\frac{b^2-a^2}{b^2}}$
14	Sum of focal distances (focal radii) of a point P on ellipse	$SP+S'P=2a$	$SP+S'P=2b$	$SP+S'P=2a$	$SP+S'P=2b$
15	Distance between the foci SS'	$2ae$	$2be$	$2ae$	$2be$
16	Distance between vertices	$AA'=2a$	$BB'=2b$	$AA'=2a$	$BB'=2b$
17	Distance between directrices ZZ'	$2a/e$	$2b/e$	$2a/e$	$2b/e$

- If PSP^1 is the focal chord of the ellipse and SL be the semi latusrectum then $\frac{1}{SP}+\frac{1}{S^1P}=\frac{2}{SL}$
(The semilatusrectum is the hormonic mean of the segments of the focal chord)

W.E-4: ***Ratio of the greatest and least focal distances of a point on the ellipse*** $4x^2+9y^2=36$ ***is***

Sol: Ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1\,(a>b)$

$$a^2=9, b^2=4, e=\sqrt{\frac{a^2-b^2}{a^2}}=\sqrt{\frac{9-4}{9}}=\frac{\sqrt{5}}{3}$$

Ratio $=1+e:1-e=1+\frac{\sqrt{5}}{3}:1-\frac{\sqrt{5}}{3}$

$=3+\sqrt{5}:3-\sqrt{5}$

W.E-5 : ***The distances from the foci of the point*** $P(x_1,y_1)$ ***on the ellipse*** $\frac{x^2}{16}+\frac{y^2}{9}=1$ ***are***

Sol: Given ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$, $a^2=16, b^2=9$

$$e=\sqrt{\frac{a^2-b^2}{a^2}}=\sqrt{\frac{7}{16}}=\frac{\sqrt{7}}{4} \qquad (a>b)$$

the focal distances $=a\pm ex_1=4\pm\frac{\sqrt{7}}{4}x_1$

W.E-6: ***If PSQ is a focal chord of the ellipse*** $16x^2+25y^2=400$ ***such that SP=8, then SQ=***

Sol: Given ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1, (a>b)$

$$\frac{1}{SP}+\frac{1}{SQ}=\frac{2}{SL}$$

$$\frac{1}{SP}+\frac{1}{SQ}=\frac{2a}{b^2}\Rightarrow\frac{1}{8}+\frac{1}{SQ}=\frac{2(5)}{16}\Rightarrow SQ=2$$

W.E-7 : ***If p,q are the segments of a focal chord of an ellipse*** $b^2x^2+a^2y^2=a^2b^2$***, then condition is***

Sol: If PSQ is a focal chord of ellipse then

$$\frac{1}{SP}+\frac{1}{SQ}=\frac{2}{SL}\Rightarrow\frac{1}{p}+\frac{1}{q}=\frac{2a}{b^2}$$

$$\Rightarrow\frac{p+q}{pq}=\frac{2a}{b^2}\Rightarrow\boxed{b^2(p+q)=2a\,pq}$$

W.E-8: ***If S and S¹ are the foci of the ellipse*** $\frac{x^2}{25}+\frac{y^2}{16}=1$ ***and P is any point on it then the range of*** $SP\times S^1P$ ***is***

Sol: Let $P(5\cos\theta, 4\sin\theta)$ be any point on the ellipse

Then $SP=5+5e\cos\theta$ and $S^1P=5-5e\cos\theta$

$\therefore SP\times S^1P=25-25e^2\cos^2\theta=16+9\sin^2\theta$

$\left(\because e=\frac{3}{5}\right)$

$16+9\sin^2\theta=f(\theta)$ (say) $\Rightarrow 16\le f(\theta)\le 25$

Notation : $S\equiv\frac{x^2}{a^2}+\frac{y^2}{b^2}-1$; $S_1\equiv\frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1$

$S_{12}\equiv\frac{x_1x_2}{a^2}+\frac{y_1y_2}{b^2}-1$; $S_{11}\equiv\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1$

- **Position of a point :** Let $P(x_1, y_1)$ be a point and $S=0$ is an ellipse. P lies on the ellipse iff $S_{11}=0$
P lies inside the ellipse iff $S_{11}<0$
P lies outside the ellipse iff $S_{11}>0$
- **Eccentric Angle :** Let P (x,y) be a point on the ellipse with centre C. let N be the foot of the perpendicular of P on the major axis. Let NP meets the auxiliary circle at P^1. Then $\angle NCP^1$ is called eccentric angle of P. The point P^1 is called corresponding point of P and its range is $[0, 2\pi)$
- If a circle cuts an ellipse in four distinct points then the sum of their eccentric angles is an even multiple of π radians, i.e., $\theta_1+\theta_2+\theta_3+\theta_4=2n\pi$
- S, S^1 are the foci of an ellipse, then the normal at any point P on the ellipse is the internal angular bisector of $\lfloor S'PS$ and tangent at P is external angular bisector of $\lfloor S'PS$
- At most four normals can be drawn from a point to the ellipse and the sum of the eccentric angles of the feet of the normals to an ellipse through a point is an odd multiple of π radians, i.e., $\theta_1+\theta_2+\theta_3+\theta_4=(2n+1)\pi$

- **Parametric Coordinates :**

$x = a\cos\theta ; y = b\sin\theta$ are called parametric equations of the ellipse S = 0. 'θ' is parameter and $\theta \in [0, 2\pi)$

- Any point on the ellipse S = 0 is $(a\cos\theta, b\sin\theta)$ and it is called point θ.

Note: If centre is (h,k) then any point

$p = (h + a\cos\theta, k + b\sin\theta)$

- **Equation of the Chord :** Equation of the chord joining the points 'α' and 'β' on the ellipse S = 0 is $\frac{x}{a}\cos\left(\frac{\alpha+\beta}{2}\right)+\frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right)$

W.E-9: ***If the chord joining points $P(\alpha)$ and $Q(\beta)$ on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ subtends a right angle at the vertex A(a,0), then $\tan\left(\frac{\alpha}{2}\right)\tan\left(\frac{\beta}{2}\right)$ is***

Sol: $m_{AP} \times m_{QA} = -1$

$$\Rightarrow \frac{b\sin\alpha}{a\cos\alpha - a} \times \frac{b\sin\beta}{a\cos\beta - a} = -1$$

$$\Rightarrow \tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{-b^2}{a^2}$$

- If α, β are the eccentric angles of the extremities of a focal chord (through S(ae, 0)) of an ellipse S = 0 (a>b), then

  - $\cos\left(\frac{\alpha-\beta}{2}\right) = e\cos\left(\frac{\alpha+\beta}{2}\right)$

  - $Tan\frac{\alpha}{2}Tan\frac{\beta}{2} = \frac{e-1}{e+1}$

- If the chord joining the points α and β on the ellipse S = 0 cuts the major axis at a distance 'd' units from the centre then $Tan\frac{\alpha}{2}Tan\frac{\beta}{2} = \frac{d-a}{d+a}$

W.E-10 : ***If the chord through the points whose eccentric angles are θ and ϕ on the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ passes through a focus, then the value of $\tan\left(\frac{\theta}{2}\right)\tan\left(\frac{\phi}{2}\right)$ is***

Sol: The equation of the line joining θ and ϕ is

$$\frac{x}{5}\cos\left(\frac{\theta+\phi}{2}\right)+\frac{y}{3}\sin\left(\frac{\theta+\phi}{2}\right)=\cos\left(\frac{\theta-\phi}{2}\right)$$

If it passes through the point (4, 0), then

$$\frac{4}{5}\cos\left(\frac{\theta+\phi}{2}\right)=\cos\left(\frac{\theta-\phi}{2}\right)$$

$$\Rightarrow \frac{4}{5} = \frac{\cos\left(\frac{\theta-\phi}{2}\right)}{\cos\left(\frac{\theta+\phi}{2}\right)}$$

By componendo and dividendo

$$\Rightarrow \frac{4+5}{4-5} = \frac{\cos\left(\frac{\theta-\phi}{2}\right)+\cos\left(\frac{\theta+\phi}{2}\right)}{\cos\left(\frac{\theta-\phi}{2}\right)-\cos\left(\frac{\theta+\phi}{2}\right)}$$

$$\Rightarrow \frac{9}{-1} = \frac{2\cos\frac{\theta}{2}\cos\frac{\phi}{2}}{2\sin\frac{\theta}{2}\sin\frac{\phi}{2}} \Rightarrow \tan\frac{\theta}{2}\tan\frac{\phi}{2} = -\frac{1}{9}$$

If it passes through the point (-4, 0),

then $\tan\frac{\theta}{2}\tan\frac{\phi}{2} = -9$

- **Position of a line w.r.t an ellipse :**

Consider the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ ....(1)

and the line $y = mx + c$ ....(2)

Eliminating y from (1) & (2) we get

$$\frac{x^2}{a^2}+\frac{(mx+c)^2}{b^2}=1$$

$$\Rightarrow b^2x^2 + a^2\left(m^2x^2 + c^2 + 2mcx\right) = a^2b^2$$

$$\Rightarrow \left(a^2m^2 + b^2\right)x^2 + 2a^2mcx + a^2\left(c^2 - b^2\right) = 0$$

...(3)

Equation (3) is a quadratic equation in x. The roots (say x_1 and x_2) are real and distinct, coincident or imaginary according as the discriminant of equation (3) is positive, zero or negative respectively.
Discriminant of (3) is

$$\Delta = 4a^4m^2c^2 - 4\left(a^2m^2 + b^2\right)a^2\left(c^2 - b^2\right)$$

$$= 4a^2b^2\left[a^2m^2 + b^2 - c^2\right]$$

$\Delta > 0 \Rightarrow c^2 < a^2m^2 + b^2$;

$\Delta = 0 \Rightarrow c^2 = a^2m^2 + b^2$

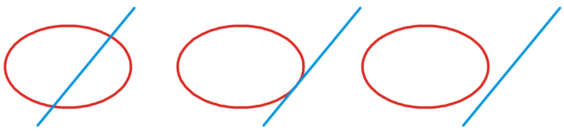
$\Delta < 0 \Rightarrow c^2 > a^2m^2 + b^2$

$\therefore$ The line $y = mx + c$

i) intersects the ellipse (1) if $c^2 < a^2m^2 + b^2$

ii) touches the ellipse (1) if $c^2 = a^2m^2 + b^2$

iii) doesnot meet the ellipse (1) if $c^2 > a^2m^2 + b^2$



- **Tangent of an ellipse:** Equation of the tangent to the ellipse S = 0 at (x_1, y_1) is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ (i.e. $S_1 = 0$)
- Equation of any tangent to the ellipse S = 0 is $y = mx \pm \sqrt{a^2m^2 + b^2}$, where m is slope of the tangent.

W.E-11 : ***The sum of the squares of the perpendiculars on any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ from two points on the minor axis each at a distance $\sqrt{a^2 - b^2}$ from the centre is***

Sol: The points on the minor axis, at a distance of $\sqrt{a^2 - b^2} = ae$ from the centre are $(0, \pm ae)$.

Lengths of the perpendiculars from $(0, \pm ae)$ to the tangent $y = mx + \sqrt{a^2m^2 + b^2}$ are

$$\frac{\left|-ae + \sqrt{a^2m^2 + b^2}\right|}{\sqrt{1 + m^2}} \text{ and } \frac{\left|ae + \sqrt{a^2m^2 + b^2}\right|}{\sqrt{1 + m^2}}$$

Sum of their squares is

$$\frac{2\left(a^2e^2 + a^2m^2 + b^2\right)}{1 + m^2} = \frac{2\left[a^2e^2 + a^2m^2 + a^2\left(1 - e^2\right)\right]}{1 + m^2} = 2a^2$$

- Two tangents can be drawn to an ellipse from an external point.
- If $y = mx + c$ is a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then the condition is $c^2 = a^2m^2 + b^2$ and point of contact is $\left(-\frac{a^2m}{c}, \frac{b^2}{c}\right)$

W.E-12 : ***The line $2x + 3y = 12$ touches the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 2$ at the point is***

Sol: Given line $2x + 3y = 12$

$$\Rightarrow y = \frac{-2x}{3} + 4, m = \frac{-2}{3}, c = 4$$

Given ellipse $\frac{x^2}{18} + \frac{y^2}{8} = 1, a^2 = 18, b^2 = 8$

point of contact $= \left(-\frac{a^2m}{c}, \frac{b^2}{c}\right) = (3, 2)$

W.E-13: ***A variable tangent to the circle $x^2 + y^2 = 1$ intersects the ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$ at points P and Q. The locus of the point of intersection of tangents to the ellipse P and Q is another ellipse. If e_1 and e_2 are the eccentricities of the two ellipses then $e_1 e_2$ is equal to***

Sol: Equation of a tangent to the circle is

$x\cos\theta + y\sin\theta = 1$

If (h, k) be a point in the locus, the equation of PQ is $hx + 2ky = 4$

As the two equations represent the same line

$h = 4\cos\theta, k = 2\sin\theta$

Locus of (h, k) is $\frac{x^2}{16} + \frac{y^2}{4} = 1$

$$e_1 = \sqrt{\frac{4-2}{4}}, e_2 = \sqrt{\frac{16-4}{16}}$$

$$e_1e_2 = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2\sqrt{2}}$$

W.E-14: ***Minimum area of the triangle by any tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with the coordinate axes is***

Sol: Equation of tangent at $(a\cos\theta, b\sin\theta)$ is

$\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1$

$\therefore$ Coordinates of P and Q are $\left(\frac{a}{\cos\theta},0\right)$ and $\left(0,\frac{b}{\sin\theta}\right)$ respectively.

$\therefore$ Area of $\Delta OPQ=\frac{1}{2}\left|\frac{a}{\cos\theta}\times\frac{b}{\sin\theta}\right|=\frac{ab}{|\sin 2\theta|}$

$\therefore$ Minimum area $=ab$

$(\max .of\ |\sin 2\theta|=1)$

- If $lx+my+n=0$ is a tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ then the condition is $a^2l^2+b^2m^2=n^2$ and point of contact is $\left(\frac{-a^2l}{n},\frac{-b^2m}{n}\right)$
- Equation of the tangent at θ to the ellipse S = 0 is $\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1$

W.E-15: ***Let d_1 and d_2 be the lengths of the perpendiculars drawn from foci S and S[1] of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ to the tangent at any point P on the ellipse. Then SP : S[1]P =***

Sol: Tangent at $P(a\cos\alpha, b\sin\alpha)$ is

$\frac{x}{a}\cos\alpha+\frac{y}{b}\sin\alpha=1$

Distance of focus $S(ae, 0)$ from this tangent is

$$d_1=\frac{|e\cos\alpha-1|}{\sqrt{\frac{\cos^2\alpha}{a^2}+\frac{\sin^2\alpha}{b^2}}}=\frac{1-e\cos\alpha}{\sqrt{\frac{\cos^2\alpha}{a^2}+\frac{\sin^2\alpha}{b^2}}}$$

Distance of focus $S^1(-ae, 0)$ from this tangent is

$$d_2=\frac{1+e\cos\alpha}{\sqrt{\frac{\cos^2\alpha}{a^2}+\frac{\sin^2\alpha}{b^2}}}\Rightarrow\frac{d_1}{d_2}=\frac{1-e\cos\alpha}{1+e\cos\alpha}$$

Now $SP=a-ae\cos\alpha$ and $S^1P=a+ae\cos\alpha$

$\Rightarrow\frac{SP}{S^1P}=\frac{1-e\cos\alpha}{1+e\cos\alpha}\Rightarrow\frac{SP}{S^1P}=\frac{d_1}{d_2}$

- Equation of tangent to the ellipse $\frac{(x-\alpha)^2}{a^2}+\frac{(y-\beta)^2}{b^2}=1$ having slope 'm' is $y-\beta=m(x-\alpha)\pm\sqrt{a^2m^2+b^2}$
- Equation of the chord of contact of (x_1, y_1) to the ellipse S=0 is $S_1=0$

W.E-16 : ***If the chords of contact of tangents from two points (x_1,y_1) and (x_2,y_2) to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are at right angles, then $\frac{x_1x_2}{y_1y_2}$ is equal to***

Sol: Chords of contact are $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$ and $\frac{xx_2}{a^2}+\frac{yy_2}{b^2}=1$; Product of slopes = -1

$\Rightarrow\left(-\frac{x_1}{a^2}.\frac{b^2}{y_1}\right)\left(-\frac{x_2}{a^2}.\frac{b^2}{y_2}\right)=-1\Rightarrow\frac{x_1x_2}{y_1y_2}=-\frac{a^4}{b^4}$

- The equation of the chord of the ellipse S=0 having (x_1,y_1) as its mid point is $S_1=S_{11}$
- The mid point of the chord $lx+my+n=0$ of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $\left(\frac{-a^2ln}{a^2l^2+b^2m^2},\frac{-b^2mn}{a^2l^2+b^2m^2}\right)$
- The area of an ellipse S=0 is πab sq.units
- The maximum area of a rectangle that can be inscribed in the ellipse S=0 is $2ab$ sq.units and the sides are $a\sqrt{2}, b\sqrt{2}$.

➢ The product of the perpendiculars drawn from the foci on any tangent of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is b^2 (if a>b), is a^2 (if a<b)

➢ The locus of the foot of the perpendicular drawn from the centre upon any tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $\left(x^2+y^2\right)^2=a^2x^2+b^2y^2$

➢ **Pair of tangents and their chord of contact** : If m_1 & m_2 are the slopes of tangents to the ellipse S = 0 drawn from (x_1, y_1) then m_1 & m_2 are satisfying the equation

$(x_1^2-a^2)m^2-2x_1y_1m+(y_1^2-b^2)=0$ and

➢ $m_1+m_2=\frac{2x_1y_1}{x_1^2-a^2}$ and $m_1 m_2=\frac{y_1^2-b^2}{x_1^2-a^2}$

➢ $m_1 - m_2=\frac{2ab\sqrt{S_{11}}}{x_1^2-a^2}$

W.E-17: ***If m_1, m_2 be the slopes of the two tangents drawn from $(1,2)$ to the ellipse $2x^2+3y^2=6$, then $m_1+m_2=$***

Sol: Ellipse $\frac{x^2}{3}+\frac{y^2}{2}=1(a>b),(x_1,y_1)=(1,2)$

$m_1+m_2=\frac{2x_1y_1}{x_1^2-a^2}=-2$

➢ If θ is the acute angle between the tangents, drawn from (x_1, y_1) to the ellipse S = 0, then

$Tan\theta=\left|\frac{2ab\sqrt{S_{11}}}{x_1^2+y_1^2-a^2-b^2}\right|$

➢ The equation of the pair of tangents to the ellipse S=0 from (x_1,y_1) is $S_1^2=S.S_{11}$

W.E-18: ***The equation of the pair of tangents drawn from the point (1, 2) to the ellipse $3x^2+2y^2=5$ is***

Sol: The combined equation of the pair of tangents drawn from (1, 2) to the ellipse $3x^2+2y^2=5$ is $S_1^2=S.S_{11}$

$\Rightarrow(3x+4y-5)^2=\left(3x^2+2y^2-5\right)(3+8-5)$

$\Rightarrow 9x^2-4y^2-24xy+40y+30x-55=0$

W.E-19 : ***Tangents are drawn from the point P(3, 4) to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ touching the ellipse at A and B. The equation of the locus of the point whose distances from the point P and the line AB are equal, is***

Sol: AB being the chord of contact of the ellipse from $P(3,4)$ has its equation is $S_1=0$

$\Rightarrow\frac{3x}{9}+\frac{4y}{4}=1\Rightarrow x+3y=3$

If (h,k) is any point on the locus, then

$\sqrt{(h-3)^2+(k-4)^2}=\left|\frac{h+3k-3}{\sqrt{1+9}}\right|$

$\Rightarrow 10\left(h^2+k^2-6h-8k+25\right)=(h+3k-3)^2$

Locus of (h,k) is

$9x^2+y^2-6xy-54x-62y+241=0$

Director Circle & Auxiliary Circle

Director Circle : The locus of point of intersection of perpendicular tangents to an ellipse is a circle concentric with the ellipse. This circle is called director circle of the ellipse.

➢ Equation of the Director circle of the ellipse S = 0 is $x^2+y^2=a^2+b^2$

W.E-20: ***Number of points on the ellipse $\frac{x^2}{50}+\frac{y^2}{20}=1$ from which pair of perpendicular tangents are drawn to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ is***

Sol: For the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$, equation of director circle is $x^2+y^2=25$. The director circle will cut the ellipse $\frac{x^2}{50}+\frac{y^2}{20}=1$ at four points. Hence, number of points is 4.

W.E-21 : ***The locus of point of intersection of the perpendicular tangents to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is***

Sol: The locus of point of intersection of the perpendicular tangents to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is a director circle and whose equation is $x^2+y^2=a^2+b^2$

$\therefore$ The required equation of director circle is $x^2+y^2=9+4 \Rightarrow x^2+y^2=13$

Note: For the ellipse $\frac{(x-\alpha)^2}{a^2}+\frac{(y-\beta)^2}{b^2}=1$, the director circle is $(x-\alpha)^2+(y-\beta)^2=a^2+b^2$

W.E-22 : ***The locus of point of intersection of orthogonal tangents to the ellipse $\frac{(x-1)^2}{16}+\frac{(y-2)^2}{9}=1$ is***

Sol: locus is director circle,

$(\alpha,\beta)=(1,2), a^2=16, b^2=9$

Equation of director circle is

$$(x-\alpha)^2+(y-\beta)^2=a^2+b^2$$

$$\Rightarrow (x-1)^2+(y-2)^2=16+9$$

$$\Rightarrow (x-1)^2+(y-2)^2=25$$

W.E-23: ***An ellipse is sliding along the coordinate axes. If the foci of the ellipse are (1, 1) and (3, 3), then area of the director circle of the ellipse (in sq. units) is***

Sol : Since x-axis and y-axis are perpendicular tangents to the ellipse. (0, 0) lies on the director circle and midpoint of foci (2, 2) is centre of the circle.

Hence, radius $=2\sqrt{2}$ $\Rightarrow$ the area is 8π units.

- **Auxiliary Circle :** The locus of the feet of the perpendiculars drawn from foci to any tangent to the ellipse is a circle concentric with the ellipse. This circle is called auxiliary circle of the ellipse.
- Equation of the auxiliary circle of the ellipse S = 0 is i) $x^2+y^2=a^2\,(a>b)$

  ii) $x^2+y^2=b^2\,(a<b)$
- The auxiliary circle of an ellipse is the circle on the major axis of the ellipse as diameter.
- **Normal :** Equation of normal at (x_1, y_1) to the ellipse S= 0 is $\frac{a^2x}{x_1}-\frac{b^2y}{y_1}=a^2-b^2$
- Equation of the normal at 'θ' to the ellipse S = 0 is $\frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2-b^2$
- If a line $y=mx+c$ be a normal to an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then the condition is $c^2=\frac{m^2\left(a^2-b^2\right)^2}{\left(a^2+b^2m^2\right)}$
- Equation of the normal to the ellipse S = 0 having slope m is $y=mx\pm\frac{m(a^2-b^2)}{\sqrt{a^2+b^2m^2}}$ is called slope form of the normal
- The condition that the line lx + my + n = 0 may be a normal to the ellipse S = 0 is $\frac{a^2}{l^2}+\frac{b^2}{m^2}=\frac{\left(a^2-b^2\right)^2}{n^2}$

W.E-24 : ***No. of normals that can be drawn at the point $(-2,3)$ to the ellipse $3x^2+2y^2=30$ are***

Sol:

$$S_{11}\equiv 3(-2)^2+2(3)^2-30$$

$$\equiv 12+18-30=0$$

$\therefore$ no. of normals = 1

W.E-25 : ***Number of normals that can be drawn from the point $(0,0)$ to the ellipse $3x^2+2y^2=30$ are***

Sol : Centre of ellipse = $(0,0)$

Number of normals from centre = 2

- The tangents at the ends of the focal chord meet on the corresponding directrix.
- If S $\equiv ax^2+2hxy+by^2+2gx+2fy+c$ and S = 0 represents an ellipse then to find the centre of the ellipse, solve the equations $\frac{\partial s}{\partial x}=0; \frac{\partial s}{\partial y}=0$

- If α,β and γ are three eccentric angles of three points on the ellipse, the normals at which are concurrent, then
$\sin(\alpha+\beta)+\sin(\beta+\gamma)+\sin(\gamma+\alpha)=0$

W.E-26 : ***If the normal at P on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ cuts the major and minor axes in Q and R respectively then PQ : PR =***

Sol: Equation of the normal at $P(x_1,y_1)$ is

$$\frac{a^2x}{x_1}-\frac{b^2y}{y_1}=a^2-b^2$$

Normal cuts major axis at $Q\left(\frac{a^2-b^2}{a^2}x_1,0\right)$ and

minor axis at $R\left(0,\frac{-a^2+b^2}{b^2}y_1\right)=R\left(0,\frac{b^2-a^2}{b^2}y_1\right)$

Now $PQ^2=\left(\frac{a^2-b^2}{a^2}x_1-x_1\right)^2+y_1^2=\frac{b^4}{a^4}x_1^2+y_1^2$

$$=\frac{b^4x_1^2+a^4y_1^2}{a^4}$$

$$PR^2=x_1^2+\left(y_1-\frac{b^2-a^2}{b^2}y_1\right)^2=x_1^2+y_1^2\frac{a^4}{b^4}$$

$$=\frac{b^4x_1^2+a^4y_1^2}{b^4};\ \frac{PQ^2}{PR^2}=\frac{b^4}{a^4}\Rightarrow\frac{PQ}{PR}=\frac{b^2}{a^2}$$

W.E-27: ***If normal at any point P to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b$ meet the axes at M and N so that $\frac{PM}{PN}=\frac{2}{3}$, then value of eccentricity e is***

Sol: $\frac{PM}{PN}=\frac{2}{3}\Rightarrow\frac{b^2}{a^2}=\frac{2}{3};$

$$e=\sqrt{1-\frac{b^2}{a^2}}=\sqrt{1-\frac{2}{3}}=\frac{1}{\sqrt{3}}$$

W.E-28: ***If the normals at $P(\theta)$ and $Q\left(\frac{\pi}{2}+\theta\right)$ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meet the major axis in G and g respectively, then $PG^2+Qg^2=$***

Sol: Normal at $P(\theta)$ is $\frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2-b^2$ ....(1)

Normal at $P\left(\frac{\pi}{2}+\theta\right)$ is

$$\frac{ax}{\cos\left(\frac{\pi}{2}+\theta\right)}-\frac{by}{\sin\left(\frac{\pi}{2}+\theta\right)}=a^2-b^2 \quad \text{(or)}$$

$$-\frac{ax}{\sin\theta}-\frac{by}{\cos\theta}=a^2-b^2 \quad ...(2)$$

Equation (1) and (2) meet the major axis at G

$\left(\frac{(a^2-b^2)\cos\theta}{a},0\right)$ and $g\left(-\frac{(a^2-b^2)\sin\theta}{a},0\right)$

Now $PG^2+Qg^2=\left(\frac{(a^2-b^2)\cos\theta}{a}-a\cos\theta\right)^2$

$$+(0-b\sin\theta)^2+\left(\frac{(a^2-b^2)\sin\theta}{a}-a\sin\theta\right)^2$$

$$+(0-b\cos\theta)^2$$

$$=\frac{(a^2-b^2)^2}{a^2}+b^2+a^2$$

$$=a^2\left(\frac{(a^2-b^2)^2}{a^4}+\frac{b^2}{a^2}+1\right)$$

$$=a^2\left(\left(1-\frac{b^2}{a^2}\right)^2+\frac{b^2}{a^2}+1\right)$$

$$=a^2\left(e^4+2-e^2\right)$$

W.E-29 : ***C is the centre of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and L is an end of a latus rectum. If the normal at L meets the major axis in G, then CG =***

Sol: Equation of normal at $L\left(ae,\frac{b^2}{a}\right)$ is

$$\frac{ax}{e}-y=a^2-b^2$$

Since it meets the major axis at G then

$$G=\left(\frac{e}{a}\left(a^2-b^2\right),0\right)\quad G=\left(-ae^3,0\right)\ \therefore CG=ae^3$$

W.E-30 : ***If the normal at $P(\theta)$ on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with focus S, meets the major axis in G, then SG=***

Sol: Equation of the normal at $P(\theta)$ is

$$\frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2e^2$$

The point where it meets the x-axis is $G=\left(ae^2\cos\theta,0\right)$ and focus $S=(ae,0)$

$$\therefore SG=\left|ae-ae^2\cos\theta\right|=ae\left(1-e\cos\theta\right)=e.SP$$

$$\therefore SG=e.SP$$

➢ **Diameter of an Ellipse :** The locus of the mid point of a system of parallel chords of an ellipse is called a diameter and passess through centre.

➢ The equation of the diameter bisecting the chords of slope m of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

$$y=-\frac{b^2}{a^2m}x$$

Proof: Let (x_1,y_1) mid point of one of the parallel chords. Then the equation of the chord is

$$S_1=S_{11}\Rightarrow\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}$$

The slope of the chord is $m\Rightarrow m=-\frac{b^2x_1}{a^2y_1}$

$\therefore$ The locus of (x_1,y_1) is $y=-\frac{b^2}{a^2m}x$

Conjugate Diameters : Two diameters of an ellipse are said to be conjugate diameters if each bisects the chords parallel to the other.

➢ Two straight lines $y=m_1x$ and $y=m_2x$ are conjugate diameters of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, if $m_1m_2=-\frac{b^2}{a^2}$

Note: In an ellipse, the major axis bisects all the chords parallel to the minor axis and vice - versa, therefore major and minor axis of an ellipse are conjugate diameters of the ellipse but they do not satisfy the condition $m_1m_2=-\frac{b^2}{a^2}$. These are the only perpendicular conjguate diameters.

W.E-31: ***If 2y=x and 3y+4x=0 are the equation of a pair of conjugate diameters of an ellipse, then the eccentricity of the ellipse is***

Sol: Slope of 2y=x is $m_1=\frac{1}{2}$ and slope of 3y+4x=0 is $m_2=\frac{-4}{3}$

$$m_1m_2=-\frac{b^2}{a^2}\Rightarrow\left(\frac{1}{2}\right)\left(-\frac{4}{3}\right)=-\frac{b^2}{a^2}\Rightarrow\frac{b^2}{a^2}=\frac{2}{3}$$

$$\therefore\quad e=\sqrt{1-\frac{b^2}{a^2}}\Rightarrow e=\sqrt{1-\frac{2}{3}}=\frac{1}{\sqrt{3}}$$

Properties :

1) The eccentric angles of the ends of a pair of conjugate diameters of an ellipse differ by a right angle.

W.E-32 : ***If θ and ϕ are eccentric angles of ends of pair of conjugate diameters of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then $\theta-\phi$ is***

Sol: Let $y=m_1x$ and $y=m_2x$ be the pair of conjugate diameters of the ellipse S=0 and let $P=\left(a\cos\theta,b\sin\theta\right)$ and $Q=\left(a\cos\phi,b\sin\phi\right)$ are the ends of these two diameters, then

$$m_1m_2=-\frac{b^2}{a^2}\Rightarrow\frac{b\sin\theta-0}{a\cos\theta-0}\times\frac{b\sin\phi-0}{a\cos\phi-0}=-\frac{b^2}{a^2}$$

$$\Rightarrow\sin\theta\sin\phi=-\cos\theta\cos\phi$$

$$\Rightarrow\cos\left(\theta-\phi\right)=0\Rightarrow\theta-\phi=\pm\frac{\pi}{2}$$

2) The sum of the squares of the two conjugate semi diameters of an ellipse is constant and is equal to the sum of the squares of the semi axes of the ellipse. $\left(i.e., \; a^2+b^2\right)$

3) The product of the focal distances of a point on the ellipse is equal to the square of the semi diameter which is conjugate to the diameter through the point.

4) The tangents at the ends of a pair of conjugate diameters of an ellipse form a parallelogram of constant area equal to the product of the axes of the ellipse (i.e., $4ab$)

- Area of the triangle formed by 3 points on an ellipse whose eccentric angles are θ, ϕ, ψ is

$$2ab\left|\sin\frac{\phi-\psi}{2}.\sin\frac{\psi-\theta}{2}.\sin\frac{\theta-\phi}{2}\right|$$

- Perpendicular from focus to any tangent and line join of centre, point of contact intersect on corrosponding directrix.
- Tangents at the ends of any chord meet on the diameter which bisects the chord.
- Two ellipses are similar if they have same eccentricity.
- Circle with focal length as diameter touches auxiliary circle
- **Equation of the ellipse referred to two perpendicular lines** : Consider the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$; Let $P(x,y)$ be any point on the ellipse

Draw perpendiculars PM and PN from P to x-axis and y-axis respectively.

Here $PM=|y|, PN=|x|$

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ can be considered as

$\frac{PN^2}{a^2}+\frac{PM^2}{b^2}=1$ i.e.,

$$\frac{(\text{perpendicular length from P to Minor Axis})^2}{(\text{L. of semi Major Axis})^2}+$$

$$\frac{(\text{perpendicular length from P to Major Axis})^2}{(\text{L. of semi Minor Axis})^2}=1$$

Instead of taking the coordinate axes as axes of the ellipse, let us take the perpendicular lines.

$L_1=a_1x+b_1y+c_1=0$

$L_2=b_1x-a_1y+c_2=0$ as axes of the ellipse

Let a, b(a>b) be the lengths of semi Major Axis and semi Minor Axis respectively.

Equation of Ellipse referred to these perpendicular lines will be $\frac{(PN)^2}{a^2}+\frac{(PM)^2}{b^2}=1$

i.e. $$\frac{\left\{\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}\right\}^2}{a^2}+\frac{\left\{\frac{b_1x-a_1y+c_2}{\sqrt{a_1^2+b_1^2}}\right\}^2}{b^2}=1 \quad ...(1)$$

Here this equation represents an ellipse for which Major axis lies along $b_1x-a_1y+c_2=0$ and Minor Axis lies along $a_1x+b_1y+c_1=0$. Centre of Ellipse will be obtained by solving

$a_1x+b_1y+c_1=0$ and $b_1x-a_1y+c_2=0$

If a < b then equation (1) represents an ellipse for which Major axis lies along $a_1x+b_1y+c_1=0$ and Minor axis lies along $b_1x-a_1y+c_2=0$.

If a > b, by solving

$a_1x+b_1y+c_1 \pm ae\sqrt{a_1^2+b_1^2}=0$ and

$b_1x-a_1y+c_2=0$ we get foci

If a < b, by solving

$b_1x-a_1y+c_2 \pm be\sqrt{a_1^2+b_1^2}=0$ and

$a_1x+b_1y+c_1=0$ we get foci

If $a>b, a_1x+b_1y+c_1 \pm \frac{a}{e}\sqrt{a_1^2+b_1^2}=0$ will be the equation of directrix.

$b_1x-a_1y+c_2 \pm \frac{b}{e}\sqrt{a_1^2+b_1^2}=0$ will be the equation of directrix.

W.E-33 : ***Find the major axis, minor axis, centre and eccentricity to the ellipse***

$$4(x-2y+1)^2+9(2x+y+2)^2=180$$

Sol: The given ellipse is

$$4(x-2y+1)^2+9(2x+y+2)^2=180$$

$$\Rightarrow \frac{(x-2y+1)^2}{45}+\frac{(2x+y+2)^2}{20}=1$$

$$\Rightarrow \frac{\left(\frac{x-2y+1}{\sqrt{1+4}}\right)^2}{3^2}+\frac{\left(\frac{2x+y+2}{\sqrt{4+1}}\right)^2}{2^2}=1$$

$$\Rightarrow \frac{X^2}{3^2}+\frac{Y^2}{2^2}=1 \qquad ....\ (1)$$

where $X=\frac{x-2y+1}{\sqrt{5}}$ and $Y=\frac{2x+y+2}{\sqrt{5}}$

From (1) it is clear that $a=3$, $b=2$ and $a>b$

$\therefore$ It follows that

i) Length of the major axis $=2\times3=6$

ii) Length of the minor axis $=2\times2=4$

iii) Equation of the major axis is $Y=0$ i.e., $2x+y+2=0$

iv) Equation of the minor axis is $X=0$ i.e., $x-2y+1=0$

v) Centre of ellipse is the point of intersection of the lines $2x+y+2=0, x-2y+1=0$ i.e., $(-1,0)$.

vi) The eccentricity e of the ellipse is

$$e=\sqrt{1-\frac{b^2}{a^2}}=\sqrt{1-\frac{4}{9}}=\sqrt{\frac{5}{9}}=\frac{\sqrt{5}}{3}$$

C.U.Q

1. If A and B are two fixed points and if the point P moves such that PA + PB = K(constant), then the locus of P is (where $K>AB$)

1) a Circle 2) a Parabola
3) an Ellipse 4) a Hyperbola

2. Let S, S^1 be the foci of an ellipse. If $\angle BSS^1=\theta$ then its eccentricity is (where 'B' is an end point of minor axis)

1) $\tan\theta$ 2) $\sin\theta$ 3) $\cos\theta$ 4) $\cot\theta$

3. If the major axis is "n" times the minor axis of the ellipse, then eccentricity is

1) $\frac{\sqrt{n-1}}{n}$ 2) $\frac{\sqrt{n-1}}{n^2}$ 3) $\frac{\sqrt{n^2-1}}{n^2}$ 4) $\frac{\sqrt{n^2-1}}{n}$

4. If α and β are the eccentric angles of the extremities of a focal chord of an ellipse, then the eccentricity of the ellipse is

1) $\frac{\cos\alpha+\cos\beta}{\cos(\alpha-\beta)}$ 2) $\frac{\sin\alpha-\sin\beta}{\sin(\alpha-\beta)}$

3) $\frac{\cos\alpha-\cos\beta}{\cos(\alpha-\beta)}$ 4) $\frac{\sin\alpha+\sin\beta}{\sin(\alpha+\beta)}$

5. The normals at a point P on the ellipse having A, A^1 as vertices and S, S^1 as foci, bisects the angle

1) $\angle A^1PA$ 2. $\angle A^1PS$ 3. $\angle S^1PS$ 4. $\angle S^1PA$

6. The distance between a focus and one end of minor axis of an ellipse is

1) Length of minor-axis 2) Length of major axis
3) distance between the foci
4) Length of semi major axis

7. A point moves so that sum of the squares of its distanes from two intersecting straight lines is constant, then the locus of p is

1) a circle 2) a parabola
3) an ellipse 4) a hyperbola

8. The orbit of the earth is an ellipse with eccentricity $\frac{1}{60}$ with the sun at one focus the major axis being approximately 186×10^6 miles in length. The longest distance of earth from the sun is

1) 9145×10^4 miles 2. 9145×10^6 miles
3) 9455×10^4 miles 4. 9455×10^6 miles

9. Let 'P' be a variable point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with foci F_1 and F_2. A is the area of the triangle PF_1F_2, then the maximum value of A is

1) $b.\sqrt{a^2-b^2}$ 2) $b.\sqrt{a^2+b^2}$
3) $a.\sqrt{a^2+b^2}$ 4) $a.\sqrt{a^2-b^2}$

10. If the normal at one end of latusrectum of the ellipse with eccentricity 'e', passes through one end of minor axis, then

1) $e^4+2e^2-1=0$ 2) $e^4-2e^2+1=0$

3) $e^4+e^2-1=0$ 4) $e^4+2e^2+2=0$

11. Tangents are drawn to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$ and the circle $x^2+y^2=a^2$ at the points where a common ordinate cuts them (on the same side of the x-axis). Then the greatest acute angle between these tangents is given by

1) $\tan^{-1}\left(\frac{a-b}{2\sqrt{ab}}\right)$ 2) $\tan^{-1}\left(\frac{a+b}{2\sqrt{ab}}\right)$

3) $\tan^{-1}\left(\frac{2ab}{\sqrt{a-b}}\right)$ 4) $\tan^{-1}\left(\frac{2ab}{\sqrt{a+b}}\right)$

12. Given the base of a triangle and the product of the tangents of half of base angles, then the locus of the vertex is

1) A Parabola 2) An Ellipse

3) A Hyperbola 4) A Circle

13. Let A and B two drawing pins, AB = 10 and let a string whose ends at A, B and length is 16. The point of pencil move on paper and fixed ends always tight. We get a curve on paper its area is

1) 36π 2) 100π

3) $16\sqrt{39}\pi$ 4) $8\sqrt{39}\pi$

14. Area of the greatest rectangle that can be inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

1) $\frac{a}{b}$ sq.unit 2) $\sqrt{ab}$ sq.unit

3) ab sq.unit 4) $2ab$ *sq.unit*

15. The normal to the curve at $P(x,y)$ meets the *x*-axis at *G* If the distance of *G* from the origin is twice the abscissa of *P*, then the curve is

1) ellipse 2) parabola

3) circle 4) hyperbola or ellipse

16. The tangent at any point P on the ellipse meets the tangents at the vertices A & A^1 of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at L and M respectively. Then AL .A^{1}M =

1) a^2 2) b^2

3) a^2+b^2 4) ab

17. The maximum distance of any normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ from the centre is

1) $a+b$ 2) $a-b$

3) a^2+b^2 4) a^2-b^2

18. The number of normals to the ellipse which passing through a focus (except major axis) are

1) 1 2) 2 3) ∞ 4) 0

19. The points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ such that the tangent at each point makes equal angles with the axes are

1. $\left[\pm\frac{a^2}{\sqrt{a^2+b^2}},\pm\frac{b^2}{\sqrt{a^2+b^2}}\right]$

2. $\left[\pm\frac{a^2}{\sqrt{a^2-b^2}},\pm\frac{b^2}{\sqrt{a^2+b^2}}\right]$

3. $\left[\pm\frac{a^2}{\sqrt{a^2-b^2}},\pm\frac{b^2}{\sqrt{a^2-b^2}}\right]$

4. $\left[\pm\frac{a^3}{\sqrt{a^2+b^2}},\pm\frac{b^3}{\sqrt{a^2+b^2}}\right]$

20. An ellipse slides between two lines at right angles to one another. The locus of its centre is

1) a parabola 2) an ellipse

3) a circle 4) Pair of straight lines

21. Two circles are given such that one is completely lying inside other without touching. The locus of the centre variable circle which touches smaller circle from outside and bigger circle from inside is

1) a circle 2) a parabola

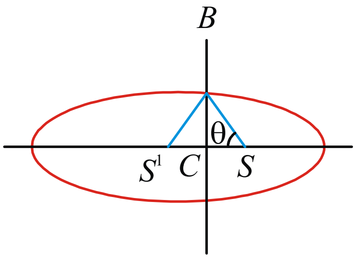
3) an ellipse 4) a hyperbola

C.U.Q - KEY

01) 3	02) 3	03) 4	04) 4	05) 3	06) 4
07) 3	08) 3	09) 1	10) 3	11) 1	12) 2
13) 4	14) 4	15) 4	16) 2	17) 2	18) 4
19) 1	20) 3	21) 3			

C.U.Q - HINTS

1. Let $A(a,0), B(-a,0)$ and $P(x,y)$
Substitute these in PA+PB=K and simplify
PA+PB=K represents ellipse if $K > 2a$
$\Rightarrow K > AB$

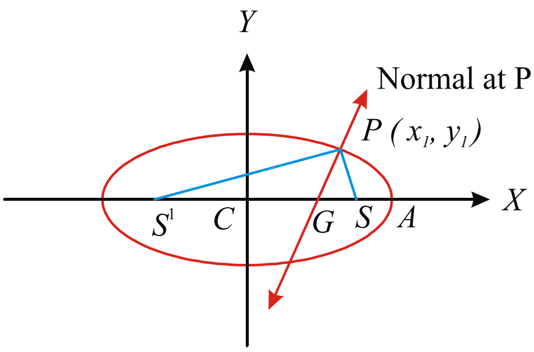
2. 

$CS = ae, SB = a \quad \cos\theta = \dfrac{ae}{a}$

3. $\cancel{2}a = n(\cancel{2}b) \Rightarrow n = \dfrac{a}{b}$

$\therefore e = \sqrt{\dfrac{a^2-b^2}{a^2}} = \sqrt{1-\dfrac{b^2}{a^2}} = \sqrt{1-\dfrac{1}{n^2}} = \dfrac{\sqrt{n^2-1}}{n}$

4. $\dfrac{x}{a}\cos\left(\dfrac{\alpha+\beta}{2}\right)+\dfrac{y}{b}\sin\left(\dfrac{\alpha+\beta}{2}\right)=\cos\left(\dfrac{\alpha-\beta}{2}\right)$
passes through (ae, 0)

5. 

Equation of normal to ellipse S=0 at $P(x_1,y_1)$ is

$y - y_1 = \dfrac{a^2 y_1}{b^2 x_1}(x-x_1)$ .....(1)

Putting $y=0$ (x-axis) in normal, we get G

$(1) \Rightarrow -y_1 = \dfrac{a^2 y_1}{b^2 x_1}(x-x_1)$ where

$b^2 = a^2(1-e^2) \Rightarrow x = e^2 x_1 \Rightarrow CG = e^2 x_1$,
where C is centre

$\therefore SG = CS - CG = ae - e^2 x_1 = e(a - ex_1) = eSP$

$\Rightarrow SG = eSP$; Similarly $S^1G = eS^1P$

$\dfrac{SG}{S^1G} = \dfrac{SP}{S^1P} \Rightarrow$ The normal at P drawn to the ellipse bisects the internal angle $\underline{|SPS^1}$

6. $S(ae,0)\, B(0,b)$

$SB = \sqrt{a^2e^2+b^2} = \sqrt{a^2e^2+a^2-a^2e^2} = a$

7. $L_1 = x = 0, L_2 = x-y=0, x^2+\left(\dfrac{x-y}{\sqrt{2}}\right)^2 = k$

$3x^2-2xy+y^2-2k=0$; $h^2-ab=-2<0$

8. $2a = 186\times10^6$, $e=\dfrac{1}{60}$ $SA^{/} = a+ae = a(1+e)$

9. For maximum of A, P=B

area $A = \dfrac{1}{2}(SS^{/}).CB \;= \dfrac{1}{2}(2ae).b$

10. $\dfrac{a^2x}{x_1}-\dfrac{b^2y}{y_1} = a^2-b^2$, one end of latus recetum $\left(ae,\dfrac{b^2}{a}\right) = (x_1,y_1)$, put (0,b) in the normal equation

11. Slope of the tangent to the ellipse at $P(a\cos\theta, b\sin\theta)$ is $m_1 = \dfrac{-b}{a}\cot\theta$.
Slope of the tangent to the circle at $Q(a\cos\theta, a\sin\theta)$ is $m_2 = -\cot\theta$

angle between two tangents $\tan\theta = \left|\dfrac{m_1-m_2}{1+m_1m_2}\right|$

$= \left|\dfrac{\cot\theta\left(1-\dfrac{b}{a}\right)}{1+\dfrac{b}{a}\cot^2\theta}\right| = \left|\dfrac{a-b}{a\tan\theta+b\cot\theta}\right|$

$\geq \left|\dfrac{a-b}{2\sqrt{ab}}\right|$

$\left(\because AM \ge GM \Rightarrow \dfrac{a\tan\theta + b\cot\theta}{2} \ge \sqrt{ab}\right.$

$\left.\Rightarrow a\tan\theta + b\cot\theta \ge 2\sqrt{ab}\right)$

12. Let the triangle be ABC with base $BC = a$

$\tan\dfrac{B}{2} = \sqrt{\dfrac{(s-c)(s-a)}{s(s-b)}}$, $\tan\dfrac{C}{2} = \sqrt{\dfrac{(s-a)(s-b)}{s(s-c)}}$,

$\tan\dfrac{B}{2}.\tan\dfrac{C}{2} = \dfrac{s-a}{s}$ If $\tan\dfrac{B}{2}\tan\dfrac{C}{2} = \lambda^2$, then

$\dfrac{s-a}{s} = \dfrac{\lambda^2}{1}, \dfrac{2s-a}{a} = \dfrac{1+\lambda^2}{1-\lambda^2}$ or

$b + c = a\left(\dfrac{1+\lambda^2}{1-\lambda^2}\right)$, constant

Hence the locus of vertex is an ellipse whose foci are B and C.

13. $SS^1 = AB = 2ae = 10 \Rightarrow ae = 5$

$SP + S^1P = 2a = 16 \Rightarrow a = 8$

$b^2 = a^2 - a^2e^2 = 64 - 25 = 39$

Area of ellipse $= \pi ab = \pi 8\sqrt{39}$

14. Let the vertices of a rectangle are $(\pm a\cos\theta, \pm b\sin\theta)$

area of rectangle $= 2a\cos\theta \times 2b\sin\theta$

$= 2ab\sin 2\theta$

Maximum area $= 2ab[\sin 2\theta = 1]$

15. Equation of the normal at (x, y) is $Y - y = -\dfrac{dx}{dy}(X - x)$ which meets the x-axis at

$G\left(0, x + y\dfrac{dy}{dx}\right)$, then $x + y\dfrac{dy}{dx} = \pm 2x$

$\Rightarrow x + y\dfrac{dy}{dx} = 2x \Rightarrow y\,dy = x\,dx \Rightarrow x^2 - y^2 = c$

or $y\,dy = -3x\,dx \Rightarrow 3x^2 + y^2 = c$

Thus the curve is either hyperbola or ellipse

16. $A = (a, 0), A^1 = (-a, 0)$

Tangent at 'θ', $\dfrac{x}{a}\cos\theta + \dfrac{y}{b}\sin\theta = 1$ meets $x = \pm a$ at L and M

$L = \left(a, \dfrac{b(1-\cos\theta)}{\sin\theta}\right), M = \left(-a, \dfrac{b(1+\cos\theta)}{\sin\theta}\right)$

$\Rightarrow AL.A^1M = b^2$

17. Equation of normal to the ellipse $S = 0$ at 'θ' is

$\dfrac{ax}{\cos\theta} - \dfrac{by}{\sin\theta} = a^2 - b^2$

Max. distance = perpendicular distance from C(0,0)

to normal $= \dfrac{|a^2 - b^2|}{\sqrt{a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta}}$

This is max. if denominater is min.

[min. value of $a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta$ is $(a+b)^2$]

$\therefore$ max. distance $= \dfrac{|a^2 - b^2|}{a+b} = (a - b)$

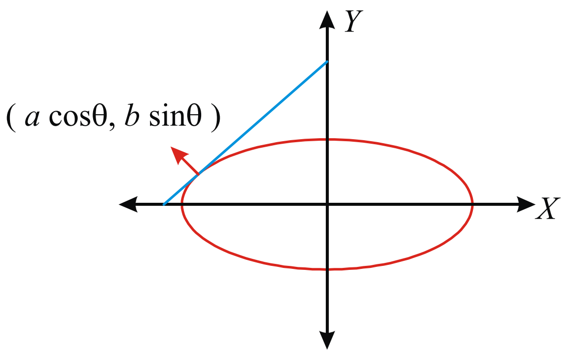
18. Normal at 'θ' $\dfrac{ax}{\cos\theta} - \dfrac{by}{\sin\theta} = a^2 - b^2$

passes through $(ae, 0)$

$\Rightarrow \dfrac{a^2e}{\cos\theta} = a^2 - b^2 \Rightarrow \cos\theta = \dfrac{1}{e}, e < 1$

$\therefore$ no normal passes through foci

19. Let P be $(a\cos\theta, b\sin\theta)$, then the equation of tangent at P is $\dfrac{x}{a}\cos\theta + \dfrac{y}{b}\sin\theta = 1$



Slope of tangent $= \dfrac{-\cos\theta}{\sin\theta}\dfrac{b}{a} = \pm\tan 45^0 = \pm 1$

$\Rightarrow \cot\theta = \pm\dfrac{a}{b} \Rightarrow \cos\theta = \pm\dfrac{a}{\sqrt{a^2 + b^2}}$

and $\sin\theta = \pm\dfrac{b}{\sqrt{a^2 + b^2}}$

$\therefore$ Coordinates of the required points are

$$\left[\pm\frac{a^2}{\sqrt{a^2+b^2}},\pm\frac{b^2}{\sqrt{a^2+b^2}}\right]$$

20. Let the two given lines be taken as the co-ordinates axes. Let $C(\alpha,\beta)$ be the centre of the ellipse in any position. Here the position of centre C changes as the ellipse slides. Let a and b be the semi major and semi-minor axes of the ellipse. Equation of the director circle of the ellipse is

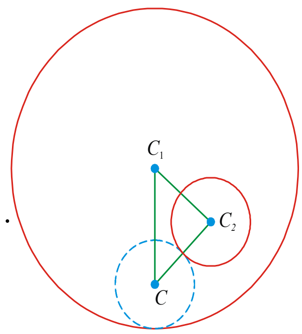
$(x-\alpha)^2+(y-\beta)^2=a^2+b^2$ ....(1)

Since OX and OY are mutually perpendicular then (1) passes through (0, 0)

$(1)\Rightarrow \alpha^2+\beta^2=a^2+b^2$

Hence, locus of $C(\alpha,\beta)$ is $x^2+y^2=a^2+b^2$

21



In the figure, circle with hard line are given circles with centres C_1 and C_2 and rdius r_1 and r_2 respectively. Let the circle with dotted line is a variable circle which touches given two circles as explained in the question which has centre C and radius r. Now $CC_2=r+r_2$ and $CC_1=r_1-r$

Hence, $CC_1+CC_2=r_1+r_2=(\text{constant})$

LEVEL-I (C.W)

STANDARD FORM

1. **The equation $\frac{x^2}{10-a}+\frac{y^2}{4-a}=1$ represents an ellipse if**

1) a<4 2) a>4
3) 4<a<10 4) a>10

2. **For an ellipse with eccentricity $\frac{1}{2}$, the centre is at the origin. If one directrix is x = 4, then the equation of the ellipse is**

1) $3x^2+4y^2=1$ 2) $3x^2+4y^2=12$
3) $4x^2+3y^2=1$ 4) $4x^2+3y^2=12$

3. **An ellipse with foci (-1,1) and (1,1) passes through (0,0) then its equation is**

1) $x^2+2y^2-8y=0$ 2) $x^2+2y^2+4y=0$
3) $x^2+2y^2+8y=0$ 4) $x^2+2y^2-4y=0$

4. **Given two fixed points A and B and AB= 6. Then simplest form of the equation to the locus of P such that PA + PB = 8 is**

1) $\frac{x^2}{16}+\frac{y^2}{7}=1$ 2) $\frac{x^2}{16}+\frac{y^2}{9}=1$
3) $\frac{x^2}{9}+\frac{y^2}{16}=1$ 4) $\frac{x^2}{12}+\frac{y^2}{21}=1$

5. **Equation of the ellipse with centre(1,2), one focus at (6,2) and passing through (4,6) is**

1) $\frac{(x-1)^2}{45}+\frac{(y-2)^2}{20}=1$

2) $\frac{(x-1)^2}{35}+\frac{(y-2)^2}{15}=1$

3) $\frac{(x-1)^2}{3}+\frac{(y-2)^2}{25}=1$

4) $\frac{(x-1)^2}{25}+\frac{(y-2)^2}{15}=1$

6. **The locus of point of intersection of the lines $\frac{tx}{a}-\frac{y}{b}+t=0,\ \frac{x}{a}+\frac{ty}{b}-1=0$ is (*t* is a parameter)**

1) Parabola 2) Circle 3) Hyperbola 4) Ellipse

7. **The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and having centre at $(0,3)$ is (2013)**

1) $x^2+y^2-6y-7=0$ 2) $x^2+y^2-6y+7=0$
3) $x^2+y^2-6y-5=0$ 4) $x^2+y^2-6y+5=0$

8. Equation of the ellipse whose axes are the axes of coordinates and which passes through the point $(-3,1)$ and has eccentricity $\sqrt{\frac{2}{5}}$ is (2011)

1) $5x^2+3y^2-48=0$ 2) $3x^2+5y^2-15=0$

3) $5x^2+3y^2-32=0$ 4) $3x^2+5y^2-32=0$

9. The equation of the ellipse whose foci are $(\pm 2,0)$ and eccentricity is 1/2, is

1) $\frac{x^2}{12}+\frac{y^2}{16}=1$ 2) $\frac{x^2}{16}+\frac{y^2}{12}=1$

3) $\frac{x^2}{16}+\frac{y^2}{8}=1$ 4) $\frac{x^2}{8}+\frac{y^2}{16}=1$

ECCENTRICITY

10. The eccentricity of the ellipse $9x^2+16y^2=576$ is

1) $\frac{\sqrt{7}}{2}$ 2) $\frac{\sqrt{5}}{4}$ 3) $\frac{7}{12}$ 4) $\frac{\sqrt{7}}{4}$

11. If the length of the major axis of an ellipse is three times the length of its minor axis, then its eccentricity is

1) $1/3$ 2) $\frac{1}{\sqrt{3}}$ 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{2\sqrt{2}}{3}$

12. The latus rectum subtends a right angle at the centre of the ellipse then its eccentricity is

1) $2\sin 18^0$ 2) $2\cos 18^0$

3) $2\sin 54^0$ 4) $2\cos 54^0$

13. The eccentricity of the conic represented by $\sqrt{(x+2)^2+y^2}+\sqrt{(x-2)^2+y^2}=8$ is

1) 1/3 2) 1/2 3) 1/4 4) 1/5

14. A circle is described with minor axis of the ellipse as diameter. If the foci lie on the circle, then the eccentricity of the ellipse is

1) $\frac{1}{\sqrt{3}}$ 2) $\frac{1}{\sqrt{2}}$ 3) $\frac{1}{\sqrt{7}}$ 4) $\frac{1}{\sqrt{5}}$

15. If the angle between the lines joining the foci to an extremity of minor axis of an Ellipse is 90^0 its eccentricity is

1) $\frac{1}{2}$ 2) $\frac{\sqrt{3}}{2}$ 3) $\frac{1}{\sqrt{3}}$ 4) $\frac{1}{\sqrt{2}}$

16. Eccentricity of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ such that the line joining the foci subtends a right angle only at two points on ellipse, is

1) $\frac{1}{2}$ 2) $\frac{1}{\sqrt{2}}$ 3) $\frac{1}{\sqrt{3}}$ 4) $\frac{1}{\sqrt{5}}$

CENTRE, VERTICES, DIRECTRICES, MAJOR & MINOR AXES

17. The centre of the ellipse $\frac{(x+y-2)^2}{9}+\frac{(x-y)^2}{16}=1$ is

1) (0, 0) 2) (1, 1) 3) (1, 0) 4) (0, 1)

18. C is the centre of the Ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and S is one focus. Then the ratio of CS to semi major axis is

1) 4:5 2) 2:3 3) 3:5 4) 2:5

19. An ellipse with centre at (0,0) cuts x axis at (3,0) and (-3,0). If its $e=\frac{1}{2}$ then the length of the semi minor axis is

1) $2\sqrt{3}$ 2) $\sqrt{5}$ 3) $3\sqrt{2}$ 4) $\frac{3\sqrt{3}}{2}$

20. The tangent at P to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ cuts the major axis in T and PN is the perpendicular to x axis. If C is the centre of the ellipse then CT.CN =

1) a 2) b 3) b^2 4) a^2

FOCUS & FOCAL DISTANCES

21. If x = 2 (Cost – Sint) ; y = 3 (Cost + Sint) represents a Conic, its foci are

1) $(\pm\sqrt{10},0)$ 2) $(\pm\sqrt{13},0)$

3) $(0,\pm\sqrt{13})$ 4) $(0,\pm\sqrt{10})$

22. A conic passing through origin has its foci at (5, 12) and (24, 7). Then its eccentricity is

1) $\frac{\sqrt{386}}{38}$ 2) $\frac{\sqrt{386}}{39}$

3) $\frac{\sqrt{386}}{47}$ 4) $\frac{\sqrt{386}}{51}$

VERTICES & DIRECTRIX

23. The directrices of the Ellipse $3x^2+4y^2+12x-8y-32=0$ are
1) x = 6, x = -10 2) x = 5, x= -11
3) x = 4, x = 12 4) x = 3, x= 9

24. The foci of an ellipse are $S(-1,-1), S^1(0,-2)$ its e=1/2, then the equation of the directrix corresponding to the focus S is
1) $x-y+3=0$ 2) $x-y+7=0$
3) $x-y+5=0$ 4) $x-y+4=0$

25. PSP[1] is focal chord of the ellipse $4x^2+9y^2=36$. If SP=4 then SP[1]
1) $\frac{2}{3}$ 2) $\frac{3}{5}$ 3) $\frac{4}{3}$ 4) $\frac{4}{5}$

26. The distances from the foci to a points $P(x_1, y_1)$ on the ellipse $\frac{x^2}{9}+\frac{y^2}{25}=1$ are
1) $4\pm\frac{2}{3}y_1$ 2) $5\pm\frac{4}{5}y_1$
3) $5\pm\frac{4}{5}x_1$ 4) $4\pm\frac{2}{3}x_1$

27. A focal chord perpendicular to major axis of the ellipse $9x^2+5y^2=45$ cuts the curve at Pand Q then length of PQ is
1) $\frac{10}{3}$ 2) $\frac{18}{\sqrt{5}}$ 3) 6 4) $2\sqrt{5}$

28. If F_1 and F_2 are the feet of perpendiculars from foci S_1 and S_2 of an ellipse $\frac{x^2}{5}+\frac{y^2}{3}=1$ on the tangent at any point P on the ellipse then $(S_1F_1)(S_2F_2)=$
1) 8 2) 5 3) 3 4) 9

LENGTH OF LATUS RECTUM

29. A point moves so that its distance from the point (2,0) is always 1/ 3 of its distance from the line x – 18 = 0. If the locus of the point is a conic, its length of latusrectum is
1) 16/ 3 2) 32/ 3
3) 8/3 4) 15/4

PARAMETRIC CO-ORDINATES

30. The distance of a point on the ellipse $x^2+3y^2=6$ from the centre is 2 units. Then the eccentric angle of the point is
1) $\frac{7\pi}{4}$ 2) $\frac{3\pi}{5}$ 3) $\frac{11\pi}{4}$ 4) $\frac{13\pi}{4}$

31. If $\frac{x}{a}+\frac{y}{b}=\sqrt{2}$ touches the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ then the eccentric angle θ of the point of contact is equal to
1) 0^0 2) 90^0 3) 45^0 4) 60^0

TANGENT & NORMAL

32. The point of intersection of the two tangents to the ellipse $2x^2+3y^2=6$ at the ends of latus rectum is
1) (3,0) 2) (7/2, 0) 3) (9/2, 0) 4) (4,0)

33. The product of the perpendiculars from the two foci of the ellipse $\frac{x^2}{9}+\frac{y^2}{25}=1$ on the tangent at any point on the ellipse is
1) 8 2) 16 3) 12 4) 9

34. The point in the first quadrant of the ellipse $\frac{x^2}{25}+\frac{y^2}{144}=1$ at which the tangent makes equal angles with the axes is
1) $\left(2,\frac{144}{3}\right)$ 2) $\left(\frac{25}{13},\frac{144}{13}\right)$
3) $\left(-\frac{25}{13},\frac{144}{13}\right)$ 4) $\left(-\frac{25}{13},-\frac{144}{13}\right)$

35. Equation of the tangent to $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (a>b) at the end of latusrectum in the first quadrant is
1) ax+ey-a=0 2) ex+y-a = 0
3) x-ey+a = 0 4) ex-2y+a = 0

36. The angle between the pair of tangents drawn from the point (1, 2) to the ellipse $3x^2+2y^2=5$ is
1) $tan^{-1}\left(\frac{12}{\sqrt{5}}\right)$ 2) $tan^{-1}\left(\frac{\sqrt{5}}{12}\right)$
3) $tan^{-1}\left(\frac{12}{5}\right)$ 4) $tan^{-1}\left(\frac{5}{12}\right)$

37. Equation of the tangent of $3x^2+4y^2=12$ parallel to $x-2y+1=0$ is

1) $x-2y+7=0$ 2) $x-2y+4=0$
3) $x-2y+5=0$ 4) $x-2y+9=0$

38. A tangent $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meets the axes at A and B. Then the locus of mid point of AB is

1) $\frac{a^2}{x^2}+\frac{b^2}{y^2}=2$ 2) $\frac{a^2}{x^2}+\frac{b^2}{y^2}=4$

3) $\frac{a^2}{x^2}+\frac{b^2}{y^2}=1$ 4) $\frac{a^2}{x^2}+\frac{b^2}{y^2}=\frac{1}{2}$

39. Equation of the normal to the ellipse $x^2+4y^2=25$ at the point whose ordinate is 2 is

1) $8x-3y-16=0$ 2) $3x-7y+15=0$
3) $8x-3y-18=0$ 4) $3x-8y-17=0$

AUXILIARY / DIRECTOR CIRCLE

40. The locus of point of Intersection of perpendicular tangents to the ellipse $\frac{(x-1)^2}{16}+\frac{(y-2)^2}{9}=1$ is

1) $(x-1)^2+(y-2)^2=25$

2) $(x-1)^2+(y-2)^2=7$

3) $(x+1)^2+(y+2)^2=25$

4) $(x+1)^2+(y+2)^2=7$

41. Equation of the auxiliary circle of the ellipse $\frac{x^2}{12}+\frac{y^2}{18}=1$ is

1) $x^2+y^2=9$ 2) $x^2+y^2=18$
3) $x^2+y^2=12$ 4) $x^2+y^2=30$

42. Tangents are drawn from any point on the circle $x^2+y^2=41$ to the Ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ then the angle between the two tangents is

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{6}$ 4) $\frac{\pi}{2}$

43. If P is a point on the Ellipse of eccentricity e and A, A^1 are the vertices and S,S^1 are the foci then $\Delta SPS^1 : \Delta APA^1$ is

1) $e^3 : 1$ 2) $e^2 : 1$ 3) $e : 1$ 4) $2e : 1$

44. A man running around a race course notes that the sum of the distances of two flag posts from him is 8 meters. The area of the path he encloses in square meters if the distance between flag posts is 4 is

1) $15\sqrt{3}\,\pi$ 2) $12\sqrt{3}\,\pi$

3) $18\sqrt{3}\,\pi$ 4) $8\sqrt{3}\pi$

LEVEL-I (C.W.) - KEY

01) 1	02) 2	03) 4	04) 1	05) 1	06) 4
07) 1	08) 1,4	09) 2	10) 4	11) 4	12) 1
13) 2	14) 2	15) 4	16) 2	17) 2	18) 3
19) 4	20) 4	21) 4	22) 1	23) 1	24) 1
25) 4	26) 2	27) 1	28) 3	29) 2	30) 1
31) 3	32) 1	33) 4	34) 2	35) 2	36) 1
37) 2	38) 2	39) 3	40) 1	41) 2	42) 4
43) 3	44) 4				

LEVEL-I (C.W.) - HINTS

1. 10 - a > 0, 4 - a > 0
2. $e=1/2, a/e=4$
3. $SS^1=2ae$ and $SP+S^1P=2a$
4. 2ae = 6; 2a = 8 ; $e=\frac{6}{8}=\frac{3}{4}$; $b^2=16-9=7$

   $\frac{x^2}{16}+\frac{y^2}{7}=1$
5. Verify (4,6)
6. $t\left(\frac{x}{a}+1\right)=\frac{y}{b}$ $\frac{x}{a}-1=\frac{-ty}{b}$

   $t\left(\frac{x^2}{a^2}-1\right)=-t\frac{y^2}{b^2}\Rightarrow$ $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$
7. Foci $=(\pm ae,0)=(\pm\sqrt{7},0)$; radius $=4$

   equation of circle having centre (0, 3) and radius 4 is $x^2+(y-3)^2=16$
8. Verification, substitute (-3, 1) in options
9. $ae=2,\ e=\frac{1}{2}$ $\therefore a=4\Rightarrow a^2=16$

   $b^2=a^2\left(1-e^2\right)=12$ ellipse $\frac{x^2}{16}+\frac{y^2}{12}=1$

10. $e=\sqrt{\frac{a^2-b^2}{a^2}}$

11. $AA^1=3BB^1$

12. $Tan45^0=\frac{LS}{CS}$

13. $S'(-2,\ 0),S(2,\ 0),2a=8 \Rightarrow a=4,\ ae=2\ ,\ e=\frac{1}{2}$

14. CB = CS

15. $Tan45^0=\frac{CB}{CS}$

16. $\tan 45^0=\frac{ae}{b}\Rightarrow ae=b\quad \Rightarrow a^2e^2=b^2$

$\Rightarrow a^2e^2=a^2-a^2e^2\quad \Rightarrow e=\frac{1}{\sqrt{2}}$

17. Point of intersection of x + y - 2 = 0; x - y = 0

18. CS : $a=e$: 1

19. $a=3;\ e=\frac{1}{2};\ b^2=a^2\left(1-e^2\right)=9\left(1-\frac{1}{4}\right)$

$b=\frac{3\sqrt{3}}{2}$

20. $CT=a\sec\theta, CN=a\cos\theta$

21. $\left(\frac{x}{2}\right)^2+\left(\frac{y}{3}\right)^2=2$, foci $=\left(0,\pm be\right)$

22. $SP+S^1P=2a$; $2ae=\sqrt{19^2+5^2}$

23. $x=h\pm a/e$

24. verify distance a/e from centre

25. $\frac{1}{SP}+\frac{1}{SP^1}=\frac{2a}{b^2}$

26. focal distances (a<b)= $b\pm ey_1$

27. $\frac{2a^2}{b}=$ L.L.R

28. $\frac{x^2}{5}+\frac{y^2}{3}=1$; $\left(S_1F_1\right).\left(S_2F_2\right)=b^2=3$

29. $l=de$; $L.L.R=2de$ (or) Conic is ellipse (e<1)

SP = ePM ; $ae=2,e=\frac{1}{3},\frac{a}{e}=18$

$\Rightarrow a=6,a^2=36\quad \Rightarrow b^2=a^2\left(1-e^2\right)=32$

L.L.R $=\frac{2b^2}{a}=\frac{32}{3}$

30. $C(0,0)$, let $P\left(\sqrt{6}\cos\theta,\sqrt{2}\sin\theta\right)$ and $CP=2$

$6\cos^2\theta+2\sin^2\theta=4 => \theta=\frac{\pi}{4},\frac{3\pi}{4},\frac{7\pi}{4}...$

31. $\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1\rightarrow(1)$

$x+y\sqrt{2}=2\sqrt{2}\rightarrow(2)\qquad (1)\equiv(2)$

32. $Z=\left(\frac{a}{e},0\right)$

33. Product of perpendiculars = a^2 (a < b)

34. $-\frac{b^2x}{a^2y}=-1$

35. Tangent at $\left(ae,\frac{b^2}{a}\right)$ to the ellipse

36. $m_1+m_2=\frac{2x_1y_1}{x_1{}^2-a^2}$; $m_1m_2=\frac{y_1{}^2-b^2}{x_1{}^2-a^2}$

$Tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|$

37. m = 1/2 $\Rightarrow$ eqn. to the tangent is

$y=mx\pm\sqrt{a^2m^2+b^2}$

38. $\frac{x}{x_1}+\frac{y}{y_1}=2$ touches the ellipse

39. Let the point (x, 2) lies on ellipse $x^2+16=25\Rightarrow x=3$ point on ellipse = (3, 2) and use verification

40. $\left(x-h\right)^2+\left(y-k\right)^2=a^2+b^2$

41. $x^2+y^2=b^2$ (a < b)

42. $x^2+y^2=41$ is the director circle of the given ellipse

43. $\Delta SPS^1:\Delta APA^1\qquad SS^1b\sin\theta:AA^1b\sin\theta$

44. $SP+S^1P=8=2a\qquad 2ae=4$

area of ellipse $=\pi ab$

LEVEL-I (H.W)

STANDARD FORM

1. **The curve with parametric equations** $x = 3(\cos t + \sin t) and \;\; y = 4(\cos t - \sin t)$ **is**
   1) An Ellipse 2) A Parabola
   3) Hyperbola 4) Circle

2. **The equation of the Ellipse with one of the foci at (4, 0) and eccentricity 4/5 is**
   1) $\frac{x^2}{25}+\frac{y^2}{9}=1$ 2) $\frac{x^2}{9}+\frac{y^2}{25}=1$
   3) $\frac{x^2}{16}+\frac{y^2}{9}=1$ 4) $\frac{x^2}{9}+\frac{y^2}{36}=1$

3. **Equation of the ellipse with focus (3,–2), eccentricity $\frac{3}{4}$ and directrix 2x - y + 3 = 0 is**
   1) $44x^2 + 36xy + 71y^2 - 374x - 528y + 756 = 0$
   2) $44x^2 + 36xy + 71y^2 - 588x + 374y + 959 = 0$
   3) $44x^2 + 36xy + 71y^2 - 125x - 274y + 659 = 0$
   4) $44x^2 + 36xy + 71y^2 - 135x - 47xy + 859 = 0$

4. **Equation to the locus of the point which moves such that the sum of the distances from the points (3,9) (3,1) is 10 is**
   1) $\frac{(x-5)^2}{9}+\frac{(y-3)^2}{25}=1$
   2) $\frac{(x-3)^2}{9}+\frac{(y-5)^2}{25}=1$
   3) $\frac{x^2}{9}+\frac{y^2}{25}=1$ 4) $\frac{(x-3)^2}{36}+\frac{(y-5)^2}{49}=1$

5. **Equation of the ellipse with foci** $(\pm 4\ ,\ 0)$ **and length of latusrectum** $\frac{20}{3}$ **is**
   1) $\frac{x^2}{20}+\frac{y^2}{36}=1$ 2) $\frac{x^2}{20}+\frac{y^2}{4}=1$
   3) $\frac{x^2}{36}+\frac{y^2}{20}=1$ 4) $\frac{x^2}{24}+\frac{y^2}{8}=1$

ECCENTRICITY

6. **The eccentricity of the ellipse $25x^2 + 9y^2 - 150x - 90y + 225 = 0$ is**
   1) $\frac{3}{5}$ 2) $\frac{4}{5}$ 3) $\frac{2}{3}$ 4) $\frac{1}{3}$

7. **In an ellipse the major and minor axes are in the ratio 5 : 3. The eccentricity of the ellipse is**
   1) $\frac{3}{5}$ 2) $\frac{1}{5}$ 3) $\frac{2}{3}$ 4) $\frac{4}{5}$

8. **(-4,1),(6,1) are the vertices of an ellipse and one of the foci lies on $x-2y=2$ then the eccentricity is**
   1) $\frac{3}{5}$ 2) $\frac{4}{5}$ 3) $\frac{2}{5}$ 4) $\frac{1}{5}$

9. **S and T are the foci of an ellipse and B is an end of the minor axis. If STB is an equilateral triangle then e is**
   1) 1/4 2) 1/3 3) 1/2 4) 2/3

10. **In the ellipse distance between the foci is equal to the distance between a focus and one end of minor axis then its eccentricity is**
    1) $\frac{1}{2}$ 2) $\frac{1}{4}$ 3) $\frac{1}{3}$ 4) $\frac{1}{5}$

11. **The ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ cuts x axis at A and y axis at B and the line joining the focus S and B makes an angle $\frac{3\pi}{4}$ with x-axis. Then the eccentricity of the ellipse is**
    1) $\frac{1}{\sqrt{2}}$ 2) $\frac{1}{2}$ 3) $\frac{\sqrt{3}}{2}$ 4) $\frac{1}{3}$

12. **If the minor axis of an ellipse forms an equilateral triangle with one vertex of the ellipse then $e=$**
    1) $\sqrt{\frac{1}{2}}$ 2) $\sqrt{\frac{2}{3}}$ 3) $\sqrt{\frac{3}{4}}$ 4) $\sqrt{\frac{4}{5}}$

13. **In an ellipse , the distances between its foci is 6 and minor axis is 8. Then its eccentricity is**
    1) $\frac{1}{2}$ 2) $\frac{4}{5}$ 3) $\frac{1}{\sqrt{5}}$ 4) $\frac{3}{5}$

CENTRE,VERTICES,DIRECTRICES, MAJOR&MINOR AXES

14. **Centre of the Ellipse $5x^2 - 6xy + 5y^2 + 22x - 26y + 29 = 0$ is**
    1) (1,-2) 2) (2,-1) 3) (1,-1) 4) (-1,2)

15. "e" is the eccentricity of the ellipse $4x^2+9y^2=36$ and C is the centre and S is the focus and A is the vertex then CS : SA=

1) $3-\sqrt{5}:\sqrt{5}$ 2) $\sqrt{5}:3-\sqrt{5}$
3) $3+\sqrt{5}:\sqrt{5}$ 4) $\sqrt{5}:3+\sqrt{5}$

16. The vertices of the Ellipse $\frac{(x+1)^2}{25}+\frac{(y-3)^2}{16}=1$ are

1) (3,4), (-6,3) 2) (4,3), (-6,3)
3) (5,3), (-7,3) 4) (6,3), (-8,3)

FOCUS & FOCAL DISTANCES

17. Foci of the Ellipse $25x^2+9y^2-150x-90y+225=0$ are

1) (3,9), (3,1) 2) (4,9), (4,1)
3) (5,7), (4,7) 4) (6,9), (5,1)

18. Distance between the foci of the Ellipse $25x^2+16y^2+100x-64y-236=0$ is

1) 8 2) 12 3) 9 4) 6

VERTICES & DIRECTRIX

19. The sum of the distances of any point on the ellipse $3x^2+4y^2=24$ from its foci is

1) $8\sqrt{2}$ 2) 8 3) $16\sqrt{2}$ 4) $4\sqrt{2}$

20. The distance of the point 'θ' on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ from a focus is

1) $a(e+\cos\theta)$ 2) $a(e-\cos\theta)$
3) $a(1+e\cos\theta)$ 4) $a(1+2e\cos\theta)$

21. If $P(x,y), F_1=(3,0), F_2=(-3,0)$ and $16x^2+25y^2=400$ then $\left(\frac{PF_1+PF_2}{2}\right)=$

1) 4 2) 5 3) 8 4) 10

LENGTH OF LATUS RECTUM

22. The eccentricity of an ellipse is $\frac{\sqrt{3}}{2}$, its length of latusrectum is

1) 1/2 (Length of major axis)
2) 1/3 (Length of major axis)
3) 1/4 (length of major axis)
4) 2/3 (length of major axis)

PARAMETRIC CO-ORDINATES

23. The points on the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ whose eccentric angles differ by a right angle are

1) $(5\cos\theta,3\sin\theta),(5\sin\theta,3\cos\theta)$
2) $(5\cos\theta,3\sin\theta),(-5\sin\theta,3\cos\theta)$
3) $(5\cos\theta,-3\sin\theta),(5\sin\theta,3\cos\theta)$
4) $(25\cos\theta,-3\sin\theta),(5\sin\theta,3\cos\theta)$

24. If $x+y\sqrt{2}=2\sqrt{2}$ is a tangent to the ellipse $x^2+2y^2=4$ then the eccentric angle of the point of contact is

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

TANGENT & NORMAL

25. If the line y = x + c touches the Ellipse $2x^2+3y^2=1$ then c =

1) $\pm\sqrt{\frac{5}{6}}$ 2) $\pm\sqrt{\frac{3}{2}}$ 3) $\pm\sqrt{\frac{2}{3}}$ 4) $\pm\sqrt{\frac{6}{5}}$

26. Equation to the pair of tangents drawn from (2,-1) to the ellipse $x^2+3y^2=3$ is

1) $y^2+4xy+4x-6y-7=0$
2) $y^2-4xy-8x+5y+9=0$
3) $y^2-4xy-6x-8y+5=0$
4) $y^2+4xy+4x+6y+9=0$

27. l_1 is the tangent to $2x^2+3y^2=35$ at (4,-1) & l_2 is the tangent to $4x^2+y^2=25$ at (2,-3). The distance between l_1 & l_2 is

1) $\frac{10}{\sqrt{73}}$ 2) $\frac{60}{\sqrt{73}}$ 3) 0 4) $\frac{5}{3\sqrt{2}}$

28. Tangents are drawn through the point $(4,\sqrt{3})$ to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$, the points at which these tangents touch the ellipse are

1) (4, 0) 2) (0, 4) 3) (-4, 0) 4) (0, -4)

29. The number of values of C such that the line y=4x+C touches the ellipse $\frac{x^2}{4}+y^2=1$ is

1) 0 2) 2 3) 1 4) infinite

30. An ellipse passes through the point $(4,-1)$ and its axes are along the coordinates. If the line $x+4y-10=0$ is a tangent to it then its equation is

1) $\frac{x^2}{100}+\frac{y^2}{5}=1$ 2) $\frac{x^2}{80}+\frac{y^2}{5/4}=1$

3) $\frac{x^2}{20}+\frac{y^2}{5}=1$ 4) $\frac{x^2}{4}+y^2=5$

AUXILIARY / DIRECTOR CIRCLE

31. The radius of the director circle of the ellipse $9x^2+25y^2-18x-100y-116=0$ is

1) $\sqrt{34}$ 2) $\sqrt{29}$ 3) 5 4) 8

32. Tangents one to each of the ellipses $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $\frac{x^2}{a^2+\lambda}+\frac{y^2}{b^2+\lambda}=1$ are drawn. If the tangents meet at right angles then the locus of their point of intersection is

1. $x^2+y^2=a^2+\lambda$ 2. $x^2+y^2=b^2+\lambda$

3. $x^2+y^2=a^2+b^2+\lambda$

4. $x^2+y^2=a^2-b^2+\lambda$

33. A tangent to the ellipse $4x^2+9y^2=36$ cut by the tangent at the extremities of the major axis at T and T'. The circle on TT' as diameter passes through the point is

1. $(\pm\sqrt{5},0)$ 2. $(0,\pm\sqrt{5})$ 3. $(0,0)$ 4. $(3,2)$

34. An ellipse with foci $(2,2),(3,-5)$ passes through (6,-1) then its semi-latusrectum is

1) $\frac{7}{2}$ 2) $\frac{5}{2}$ 3) $\frac{9}{2}$ 4) $\frac{11}{2}$

35. If N is the foot of the perpendicular drawn from any point P on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$ to its major axis AA^1 then

$\frac{PN^2}{AN.A^1N}=$

1) $\frac{a^2}{b^2}$ 2) $\frac{b^2}{a^2}$ 3) $\frac{2a^2}{b^2}$ 4) $\frac{2b^2}{a^2}$

36. A tangent to $3x^2+4y^2=12$ is equally inclined with the coordinate axes. Then the perpendicular distance from the centre of the ellipse to this tangent is

1) $\sqrt{\frac{7}{2}}$ 2) $\sqrt{\frac{5}{2}}$ 3) $\sqrt{\frac{9}{2}}$ 4) $\sqrt{\frac{11}{2}}$

37. The length of the chord intercepted by the ellipse $4x^2+9y^2=1$ on the line $9y=1$ is

1) $2\sqrt{2}$ 2) $\frac{2\sqrt{2}}{3}$ 3) $\sqrt{2}$ 4) $3\sqrt{2}$

38. The radius of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and having its centre at $(0,3)$ is

1) 4 unit 2) 3 unit 3) $\sqrt{12}$ unit 4) $\frac{7}{2}$ unit

LEVEL-I (H.W.) - KEY

01) 1	02) 1	03) 2	04) 2	05) 3	06) 2
07) 4	08) 1	09) 3	10) 1	11) 1	12) 2
13) 4	14) 4	15) 2	16) 2	17) 1	18) 4
19) 4	20) 3	21) 2	22) 3	23) 2	24) 2
25) 1	26) 1	27) 1	28) 1	29) 2	30) 2
31) 1	32) 3	33) 1	34) 2	35) 2	36) 1
37) 2	38) 1				

LEVEL-I (H.W.) - HINTS

1. $\frac{x^2}{9}+\frac{y^2}{16}=2$

2. $ae=4, e=4/5$

3. $SP^2=e^2.PM^2$

4. $2b=10\ \&\ 2be=8$

5. $ae=4,\ L.L.R=20/3$

6. $e=\sqrt{\frac{b^2-a^2}{b^2}}$

7. $2a:2b=5:3$; $\frac{a}{b}=\frac{5}{3}$.; $25a^2(1-e^2)=9a^2$.

$\therefore e=\sqrt{1-\frac{9}{25}}=\frac{4}{5}$

8. 2a= 10, focus (1 + 5e, 1) lies on x - 2y = 2

9. $Tan60^0 = \frac{b}{ae}$

10. $2ae = \sqrt{a^2e^2 + b^2}$

11. $Tan\frac{\pi}{4} = \frac{b}{ae}$

12. $Tan30^0 = \frac{b}{a}$

13. $2ae = 6, 2b = 8$; $ae = 3, b = 4$

$b^2 = a^2\left(1-e^2\right) \Rightarrow a^2 = 25$; $e = \sqrt{\frac{a^2-b^2}{a^2}} = \frac{3}{5}$

14. $\left(\frac{hf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2}\right)$

15. CS : SA = ae : (a - ae)

16. $(h \pm a, k)$

17. $25(x-3)^2 + 9(y-5)^2 = 225$

$\frac{(x-3)^2}{9} + \frac{(y-5)^2}{25} = 1$;

$\left(0, \pm 5 \times \frac{4}{5}\right) = (x-3,\ y-5) = (3,1)$ & $(3,9)$

18. distance = 2be

19. $SP + S^1P = 2a$

20. $a + ex_1$

21. $P = (x,y), F_1 = (3,0), F_2 = (-3,0)$

$\frac{x^2}{25} + \frac{y^2}{16} = 1(a > b)$; $\frac{PF_1 + PF_2}{2} = \frac{2a}{2} = a = 5$

22. $L.L.R = \frac{2b^2}{a} = \frac{2a^2\left(1-e^2\right)}{a}$

23. θ and $\frac{\pi}{2} + \theta$

24. $\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1 \rightarrow (1)$

$x + y\sqrt{2} = 2\sqrt{2} \rightarrow (2)$ $\quad (1) \equiv (2)$

25. $c^2 = a^2m^2 + b^2$

26. $S_1^2 = S.S_{11}$

27. $2(4x) - 3y = 35$
$4(2x) - 3y = 25$

$d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$

28. Points lying on ellipse and $S_1 = 0$

29. Condition for tangency is $C^2=a^2m^2+b^2$
$\Rightarrow C = \pm\sqrt{65}$

30. Verification, substitute (4, -1) in options (or) Verify condition $c^2 = a^2m^2 + b^2$

31. $r = \sqrt{a^2 + b^2}$

32. $y - mx = \sqrt{a^2m^2 + b^2}$

$my + x = \sqrt{\left(a^2 + \lambda\right) + \left(b^2 + \lambda\right)m^2}$

squaring and adding.

33. Tangent at 'θ' is $\frac{x}{3}\cos\theta + \frac{y}{2}\sin\theta = 1$ Put

$x = \pm 3$, $T = \left(3, 2\tan\frac{\theta}{2}\right), T^1 = \left(-3, 2\cot\frac{\theta}{2}\right)$

34. $2ae = \sqrt{1+49} = 5\sqrt{2}$; $SP + S^1P = 2a$

35. $P = (a\cos\theta, b\sin\theta)$; $PN = b\sin\theta$

$AN = CA - CB = a - a\cos\theta$

$A^1N = CA^1 + CN = a(1+\cos\theta)$

$\frac{b^2\sin^2\theta}{a^2(1+\cos\theta)(1-\cos\theta)} = \frac{b^2}{a^2}$

36. $m = \pm 1$; $y = mx \pm \sqrt{a^2m^2 + b^2}$

37. $y = 1/9 \Rightarrow 4x^2 + 9\frac{1}{9^2} = 1$

$4x^2 = 1 - \frac{1}{9} = \frac{8}{9}$;

$x^2 = \frac{2}{9}$; $x = \pm\frac{\sqrt{2}}{3}$

length $= |x_1 - x_2| = \frac{2\sqrt{2}}{3}$

38. Foci $= (\pm ae, 0) = \left(\pm\sqrt{7}, 0\right)$;

radius $= \sqrt{7+9} = 4$

LEVEL-II (C.W)

ECCENTRICITY

1. If the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(b>a)$ and the parabola $y^2=4ax$ cut at right angles, then eccentricity of the ellipse is

1) $\frac{3}{5}$ 2) $\frac{2}{3}$ 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{1}{2}$

2. If maximum distance of any point on the ellipse $x^2+2y^2+2xy=1$ from its centre $C(0,0)$. be r, then r is equal to

1) $3+\sqrt{3}$ 2) $2+\sqrt{2}$ 3) $\frac{\sqrt{2}}{\sqrt{3-\sqrt{5}}}$ 4) $\sqrt{2-\sqrt{2}}$

3. An ellipse passing through the point $(2\sqrt{13},4)$ has its foci at $(-4,1)$ and $(4,1)$. Then its eccentricity is

1) $\frac{2}{3}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ 4) $\frac{1}{2}$

4. The length of subtangent corresponding to the point $\left(3,\frac{12}{5}\right)$ on the ellipse is 16/3. Then the eccentricity is

1) $\frac{4}{5}$ 2) $\frac{2}{3}$ 3) $\frac{1}{5}$ 4) $\frac{3}{5}$

5. The tangent drawn to the ellipse at the parametric point θ, where $\theta=\tan^{-1}2$ meets the auxiliary circle at P and Q and PQ substends a right angle at the centre of the ellipse, then eccentricity is

1) $\frac{1}{\sqrt{3}}$ 2) $\frac{1}{3}$ 3) $\sqrt{\frac{2}{3}}$ 4) $\frac{\sqrt{5}}{3}$

6. S_1,S_2 are foci of an ellipse of major axis of length 10 units and P is any point on the ellipse such that perimeter of triangle PS_1S_2 is 15. Then eccentricity of the ellipse is

1) 0.5 2) 0.25 3) 0.28 4) 0.75

7. Locus of the point which divides double ordinates of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ in the ratio $1:2$ internally is

1) $\frac{x^2}{a^2}+\frac{9y^2}{b^2}=1$ 2) $\frac{x^2}{a^2}+\frac{9y^2}{b^2}=\frac{1}{9}$

3) $\frac{9x^2}{a^2}+\frac{9y^2}{b^2}=1$ 4) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=a^2-b^2$

8. $E_1=\frac{x^2}{a^2}+\frac{y^2}{b^2}-1=0,(a>b)$ and $E_2=\frac{x^2}{k^2}+\frac{y^2}{b^2}-1=0,(k<b)$ E_2 is inscribed in E_1.If E_1 and E_2 have same eccentricities then length of minor axis of $E_2=p(LLR \text{ of } E_1)$ then p=...

1) $\frac{1}{2}$ 2) 1 3) $\frac{2}{3}$ 4) $\frac{3}{4}$

FORMS OF ELLIPSE

9. Axes are coordinate axes, S and S[1] are foci, B and B[1] are the ends of minor axis, $\underline{|SBS^1}=\sin^{-1}\left(\frac{4}{5}\right)$. Area of SBS^1B^1 is 20 sq. units., then the equation of the ellipse is

1) $\frac{x^2}{20}+\frac{y^2}{16}=1$ 2) $\frac{x^2}{25}+\frac{y^2}{16}=1$

3) $\frac{x^2}{25}+\frac{y^2}{9}=1$ 4) $\frac{x^2}{25}+\frac{y^2}{20}=1$

10. An ellipse is drawn by taking a diameter of the circle $(x-1)^2+y^2=1$ as its semiminor axis and a diameter of the circle $x^2+(y-2)^2=4$ as its semi major axs. If the centre of the ellipse is the origin and its axes are the co-ordinate axes, then the equation of the ellipse is [AIEEE-2012]

1) $4x^2+y^2=4$ 2) $4x^2+4y^2=8$

3) $4x^2+y^2=8$ 4) $x^2+4y^2=16$

11. If the line containing a focal chord of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ intersects the auxiliary circle in Q and Q[1] then SQ . SQ[1]=

1) a^2 2) b^2
3) a^4 4) b^4

12. A variable point P on the ellipse of eccentricity is joined to the foci S and S'. The eccentricity of the locus of the in centre of the triangle PSS^1 is

1) $\sqrt{\frac{2e}{1+e}}$ 2) $\sqrt{\frac{e}{1+e}}$ 3) $\sqrt{\frac{1-e}{1+e}}$ 4) $\frac{e}{2(1+e)}$

13. Each of the four inequalities given below defines a region in the xy-plane. Let P be the property for any two points (x_1, y_1) and (x_2, y_2) in the region, the point $\left(\frac{1}{2}(x_1+x_2), \frac{1}{2}(y_1+y_2)\right)$ is also in the region. Then which of them does not satisfy property P :

1) $x^2+2y^2 \le 1$ 2) $\max\{|x|,|y|\} \le 1$
3) $x^2-y^2 \le 1$ 4) $y^2-x \le 0$

14. The ellipse $x^2+4y^2=4$ is inscribed in a rectangle aligned with the coordinate axes, which is turn in inscribed in another ellipse that passes through the point $(4,0)$. Then, the equation of the ellipse is
(2009)

1) $x^2+12y^2=16$ 2) $4x^2+48y^2=48$
3) $4x^2+64y^2=48$ 4) $x^2+16y^2=16$

TANGENTS & NORMALS

15. A line touches the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the circle $x^2+y^2=r^2$. Then the slope m of the common tangent is given by $m^2=$

1) $\frac{a^2-r^2}{b^2-r^2}$ 2) $\frac{r^2-b^2}{a^2-r^2}$ 3) $\frac{r^2+b^2}{a^2-r^2}$ 4) $\frac{r^2-2b^2}{a^2-2r^2}$

16. If the tangent drawn at a point $(t^2, 2t)$ on the parabola $y^2=4x$ is same as normal drawn at $(\sqrt{5}\cos\alpha, 2\sin\alpha)$ on the ellipse $\frac{x^2}{5}+\frac{y^2}{4}=1$, then which of following is not true

1) $t=\pm\frac{1}{\sqrt{5}}$ 2) $\alpha=-\tan^{-1}2$
3) $\alpha=\tan^{-1}2$ 4) $\alpha=\tan^{-1}4$

LOCUS

17. P(θ), $D\left(\theta+\frac{\pi}{2}\right)$ are two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Then the locus of mid point of chord PD is

1) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=2$ 2) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=4$
3) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{1}{4}$ 4) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{1}{2}$

18. The sum of the eccentric angles of two points of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is 2α (constant) then the locus of point of intersection of the two tangents at these points is

1) $ay=bx\tan\alpha$ 2) $ax=by\tan\alpha$
3) $ay=bx\cot\alpha$ 4) $ax=by\cot\alpha$

PARAMETRIC

19. The eccentric angles of the ends of latusrectum of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

1) $\tan^{-1}\left(\pm\frac{b}{ae}\right)$ 2) $\sin^{-1}\left(\pm\frac{b}{ae}\right)$
3) $\cos^{-1}\left(\pm\frac{b}{ae}\right)$ 4) $\sec^{-1}\left(\pm\frac{b}{ae}\right)$

20. If the tangent at the point $\left(4\cos\phi, \frac{16}{\sqrt{11}}\sin\phi\right)$ to the ellipse $16x^2+11y^2=256$ is also a tangent to the circle $x^2+y^2-2x=15$, then the value of ϕ is

1) $\pm\frac{\pi}{2}$ 2) $\pm\frac{\pi}{4}$ 3) $\pm\frac{\pi}{3}$ 4) $\pm\frac{\pi}{6}$

21. P is a variable point on the ellipse $9x^2+16y^2=144$ with foci S and S[1]. If K is the area of the triangle SS[1]P then the maximum value of K is

1) $7\sqrt{3}$ 2) $3\sqrt{5}$ 3) $7\sqrt{5}$ 4) $3\sqrt{7}$

22. A tangent having slope – 4/3 to the ellipse $\frac{x^2}{18}+\frac{y^2}{32}=1$ meets the major and minor axes at A and B. If O is the centre of the ellipse then the area of ΔOAB is

1) 16 Sq units 2) 20 Sq. units
3) 24 Sq.Units 4) 22 Sq.Units

23. The points of intersection of the two ellipse

$x^2+2y^2-6x-12y+23=0$

$4x^2+2y^2-20x-12y+35=0$

1) lie on a circle centered at $\left(\frac{8}{3},3\right)$ and of radius $\frac{1}{3}\sqrt{\frac{47}{2}}$

2) lie on a circle centered at $\left(-\frac{8}{3},-3\right)$ and of radius $\frac{1}{3}\sqrt{\frac{47}{2}}$

3) lie on a circle centered at $(8,9)$ and of radius $\frac{1}{3}\sqrt{\frac{47}{3}}$

4) are not concyclic

24. If the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is inscribed in a rectangle whose length to breadth ratio is 2:1 then the area of the rectangle is

1) $4\frac{a^2+b^2}{7}$ 2) $4\frac{a^2+b^2}{3}$

3) $12\frac{a^2+b^2}{5}$ 4) $8\frac{a^2+b^2}{5}$

25. A bridge is in the shape of a semi ellipse. It is 400 mts long and has a maximum height 10mts at the middle point. The height of the bridge at a point distant 80 mts. From one end is

1) 4mts 2) 2mts 3) 8mts 4) 6mts.

26. S is one focus of an ellipse and P is any point on the ellipse. If the maximum and minimum values of SP are m and n respectively, then the length of semi major axis is

1) AM of m,n 2) G.M. of m,n
3) HM of m,n 4) AGP of m,n

27. If F_1 and F_2 are the feet of the perpendiculars from the foci S_1 and S_2 of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ on the tangent at any point P on the ellipse then $S_1F_1+S_2F_2$

1. $=5$ 2. $=16$ 3. ≥ 8 4. <8

LEVEL-II (C.W.) - KEY

01) 3	02) 3	03) 4	04) 1	05) 4	06) 1
07) 1	08) 2	09) 4	10) 4	11) 2	12) 1
13) 3	14) 1	15) 2	16) 4	17) 4	18) 1
19) 1	20) 3	21) 4	22) 3	23) 1	24) 4
25) 3	26) 1	27) 3			

LEVEL-II (C.W.) - HINTS

1. $-\frac{b^2x}{a^2y}.\frac{2a}{y}=-1$

2. Here centre of the ellipse is $(0,0)$

Let $P(r\cos\theta, r\sin\theta)$ be any point on the given ellipse then

$r^2\cos^2\theta+2r^2\sin^2\theta+2r^2\sin\theta\cos\theta=1$

$\Rightarrow r^2=\frac{1}{\cos^2\theta+2\sin^2\theta+\sin 2\theta}$

$=\frac{1}{\sin^2\theta+1+\sin 2\theta}$ $=\frac{2}{1-\cos 2\theta+2+2\sin 2\theta}$

$=\frac{2}{3-\cos 2\theta+2\sin 2\theta}$ $\Rightarrow r_{\max}=\frac{\sqrt{2}}{\sqrt{3-\sqrt{5}}}$

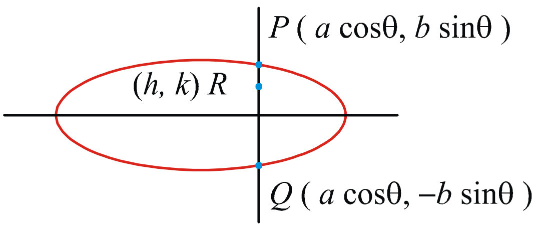
3. $SS^1 = 2ae$; $SP + S^1P = 2a$

4. Length of subtangent is $\left|\frac{y}{m}\right| = 16/3$

5. $\left(1+\sin^2\theta\right)^{-\frac{1}{2}}$

6. $S_1P + S_2P + S_1S_2 = 15 \Rightarrow 2a + 2ae = 15$

$\Rightarrow 2a(1+e) = 15$; $1+e = \frac{3}{2}$; $e = 0.5$

7. 

Let $P(a\cos\theta, bsin\theta), Q(a\cos\theta, -b\sin\theta)$

$PQ : RQ = 1:2$

There fore $h = a\cos\theta \Rightarrow \cos\theta = \frac{h}{a}$

and $k = \frac{b}{3}\sin\theta \Rightarrow \sin\theta = \frac{3k}{b}$

On squaring and adding Eqs. (i) and (ii), we get

$\frac{x^2}{a^2} + \frac{9y^2}{b^2} = 1$

8. Minor axis of E_2=L.L.R of E_1

9. Verify area of $SBS^1B^1 = 20$

10. Semi minor axis b=2 ; Semi major axis a =4

Equation of ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\Rightarrow \frac{x^2}{16} + \frac{y^2}{4} = 1 \Rightarrow x^2 + 4y^2 = 16$

11. $SQ.SQ^1 = b^2$; Auxiliary circle is $x^2 + y^2 = a^2$

Focus of ellipse $S(ae, 0)$

$\therefore SQ.SQ^1 = |S_{11}| = |x_1^2 + y_1^2 - a^2| = |a^2e^2 + 0 - a^2| = b^2$

12. Let ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ Foci $S(ae, 0)$ and $S^1(-ae, 0)$

Let $P(a\cos\theta, b\sin\theta)$ $SP(a(1-e\cos\theta))$

$S^1P = (a(1+e\cos\theta))$; $SS^1 = 2ae$

Let (h,k) be the incentre of triangle PSS^1

Then by applying the formula

$h = \frac{ax_1 + bx_2 + cx_3}{a+b+c}$

$k = \frac{ay_1 + by_2 + cy_3}{a+b+c}$ and eliminating θ will get

$\frac{h^2}{a^2e^2} + \frac{(1+e^2)k^2}{b^2e^2}$ then calculating eccentricity

will get $E = \sqrt{\frac{2e}{1+e}}$

13. Region (1), (2) and (4) are respectively inside of ellipse, square and parabola and midpoint of any two points in each region also lies in the region. But (3) represents hyperbola where midpoint of two points on this curve can be outside curve also. So, (3) does not satisfy property P.

14. The given ellipse is $x^2 + 4y^2 = 4 \Rightarrow \frac{x^2}{4} + \frac{y^2}{1} = 1$

$\Rightarrow$ the point A, the corner of the rectangle in 1st quadrant, is (2,1)

Again the ellipse circumscribing the rectangle passes through the point (4,0), so its equation is

$\frac{x^2}{16} + \frac{y^2}{b^2} = 1$

$A(2,1)$ lies on the above ellipse

$\Rightarrow \frac{x^2}{16} + \frac{y^2}{b^2} = 1 \frac{1}{b^2} = 1 - \frac{1}{4} = \frac{3}{4} \Rightarrow b^2 = \frac{4}{3}$

Thus the equation to the desired ellipse is

$\frac{x^2}{16} + \frac{3}{4}y^2 = 1 \Rightarrow x^2 + 12y^2 = 16$

15. Common tangent is $y=mx+c$

tangent to ellipse, $c^2 = a^2m^2 + b^2$ ....(1)

tangent to circle, $c^2 = r^2(1+m^2)$ ....(2)

eliminate c^2 from (1) and (2)

$a^2m^2 + b^2 = r^2 + r^2m^2 \Rightarrow m^2 = \frac{r^2 - b^2}{a^2 - r^2}$

16. Equation of tangent to $y^2 = 4x$ at $(t^2, 2t)$ is

$x = ty - t^2$ ....(A)

Equation of normal to ellipse $\frac{x^2}{5} + \frac{y^2}{20} = 1$ at $(\sqrt{5}\cos\alpha, 2\sin\alpha)$ is

$\sqrt{5}\sec\alpha x - 2y\,\mathrm{cosec}\,\alpha = 1$ ....(B)

Given (A) & (B) $\Rightarrow \sqrt{5}\sec\alpha = \frac{2\,\mathrm{cosec}\,\alpha}{t} = -\frac{1}{t^2}$

$\Rightarrow \cos\alpha = -\sqrt{5}t^2$ & $\sin\alpha = -2t$

$\Rightarrow \cos^2\alpha + \sin^2\alpha = 5t^4 + 4t^2 = 1 \Rightarrow t^2 = \frac{1}{5}$

$t^2 = \frac{1}{5}$

17. Mid point of PD = (x,y)

18. Two points $(a\cos\theta_1, b\sin\theta_1)$

$and (a\cos\theta_2, b\sin\theta_2) where\ \theta_1 + \theta_2 = 2\alpha$

$$x = \frac{a\cos\alpha}{\cos\left(\frac{\theta_1 - \theta_2}{2}\right)}, y = \frac{b\sin\alpha}{\cos\left(\frac{\theta_1 - \theta_2}{2}\right)}$$

19. Point θ is $\left(\mathrm{ae}, \pm \mathrm{b}/\mathrm{a}\right)$

20. The equation of the tangent at $\left(4\cos\phi, \frac{16}{\sqrt{11}}\sin\phi\right)$ to the ellipse $16x^2 + 11y^2 = 256$ is

$$16x(4\cos\phi) + 11y\left(\frac{16}{\sqrt{11}}\sin\phi\right) = 256$$

$$\Rightarrow 4x\cos\phi + \left(\sqrt{11}\sin\phi\right)y = 16$$

This touches the circle $(x-1)^2 + y^2 = 4^2$

$\therefore r = d$

$$\left|\frac{4\cos\phi - 16}{\sqrt{16\cos^2\phi + 11\sin^2\phi}}\right| = 4$$

$$\Rightarrow (\cos\phi - 4)^2 = 16\cos^2\phi + 11\sin^2\phi$$

$$\Rightarrow 15\cos^2\phi + 11\sin^2\phi + 8\cos\phi - 16 = 0$$

$$\Rightarrow 4\cos^2\phi + 8\cos\phi - 5 = 0$$

$$\Rightarrow (2\cos\phi - 1)(2\cos\phi + 5) = 0 \Rightarrow \cos\phi = \frac{1}{2}$$

$$\therefore \phi = \pm\frac{\pi}{3}$$

21. $\frac{1}{2}SS^1\,\mathrm{b}\sin\theta = \mathrm{abe}$

22. $y = -\frac{4}{3}x + \sqrt{18\left(-\frac{4}{3}\right)^2 + 32}$

23. If $S_1 = 0$ and $S_2 = 0$ are the equations, Then $\lambda S_1 + S_2 = 0$ is a second degree curve passing through the points of intersection of $S_1 = 0$ and $S_2 = 0$

$$\Rightarrow (\lambda + 4)x^2 + 2(\lambda + 1)y^2 - 2(3\lambda + 10)x$$
$$-12(\lambda + 1)y + (23\lambda + 35) = 0$$

For it to be a circle, choose λ such that the coefficients of x^2 and y^2 are equal: $\therefore \lambda = 2$

This gives the equaiton of the circle as $6(x^2 + y^2) - 32x - 36y + 81 = 0\{u\ \sin\ g(1)\}$

$$\Rightarrow x^2 + y^2 - \frac{16}{3}x - 6y + \frac{27}{2} = 0$$

Its centre is $C\left(\frac{8}{3}, 3\right)$ and radius is

$$r = \sqrt{\frac{64}{9} + 9 - \frac{27}{2}} = \frac{1}{3}\sqrt{\frac{47}{2}}$$

24. $\frac{2a}{2b} = \frac{2}{1} \Rightarrow a = 2b,$ area $= 2a \times 2b = 4ab = 8b^2$

option (4) $\Rightarrow 8\left(\frac{4b^2 + b^2}{5}\right) = 8b^2$

25. b = 10, 2a = 400

(120,4) lies on the ellipse $\frac{x^2}{200^2} + \frac{y^2}{10^2} = 1$

26. Max focal distance = SA^1 Min focal distance = SA

27. Now $A.M \geq G.M.$

$$\Rightarrow \frac{S_1F_1 + S_2F_2}{2} \geq \sqrt{S_1F_1.S_2F_2} \Rightarrow S_1F_1 + S_2F_2 \geq 8$$

LEVEL-II (H.W)

ECCENTRICITY

1. **S(3,4) and $S^1(9,12)$ are the foci of an ellipse and the foot of the perpendicular from S to a tangent to the ellipse is (1,-4). Then the eccentricity of the ellipse is**

1) $\frac{3}{13}$ 2) $\frac{4}{13}$ 3) $\frac{5}{13}$ 4) $\frac{7}{13}$

2. **The length of subnormal at (4,2) to an ellipse is 3. Then its eccentricity is**

1) $\frac{1}{3}$ 2) $\frac{1}{4}$ 3) $\frac{2}{3}$ 4) $\frac{1}{2}$

3. **At some point P on the ellipse, the segment SS' subtends a right angle, then its eccentricity is**

1) $e=\frac{\sqrt{2}}{2}$ 2) $e<\frac{1}{\sqrt{2}}$

3) $e>\frac{1}{\sqrt{2}}$ 4) $\frac{\sqrt{3}}{2}$

4. **An ellipse is inscribed in a rectangle and the angle between the diagonals of the rectangle is $\tan^{-1}\left(2\sqrt{2}\right)$ then the eccentricity of the ellipse is**

1) $\cot 15^0$ 2) $\cos 45^0$ 3) $\cot 60^0$ 4) $\cot 75^0$

5. **If the eccentricity of the ellipse $\frac{x^2}{a^2+1}+\frac{y^2}{a^2+2}=1$ is $\frac{1}{\sqrt{6}}$, then latus rectum of ellipse is**

1) $\frac{5}{\sqrt{6}}$ 2) $\frac{10}{\sqrt{6}}$ 3) $\frac{8}{\sqrt{6}}$ 4) $\frac{\pi}{2}$

6. **If the focal distance of an end of the minor axis of an ellipse (referred to its axes as the axes of *x* and *y*, respectively) is *k* and distance between its foci is 2*h*, then its equation**

1) $\frac{x^2}{k^2}+\frac{y^2}{k^2-h^2}=1$ 2) $\frac{x^2}{k^2}+\frac{y^2}{k^2+h^2}=1$

3) $\frac{x^2}{k^2-h^2}+\frac{y^2}{k^2+h^2}=1$ 4) $\frac{x^2}{k^2}+\frac{y^2}{k^2-h^2}=2$

7. $E_1=\frac{x^2}{25}+\frac{y^2}{9}-1=0,\ C=0$ **is a greatest circle incribed in E_1 and Greatest Ellipse E_2 is incribed in $C=0$. If the eccentricities of E_1,E_2 are equal then the minor axis of $E_2=0$ is**

1) 4 2) 5 3) $\frac{16}{5}$ 4) $\frac{18}{5}$

FORMS OF ELLIPSE

8. **The tangents from which of the following points to the ellipse $5x^2+4y^2=20$ are perependicular**

1) $\left(\sqrt{5},2\sqrt{2}\right)$ 2) $\left(2\sqrt{2},1\right)$

3) $\left(\sqrt{5},-1\right)$ 4) $\left(\sqrt{5},1\right)$

9. **Length of the chord intercepted by the ellipse $x^2+4y^2=16$ on the line $y=x\sqrt{2}+2$ is**

1) $\frac{16\sqrt{5}}{3}$ 2) $\frac{16\sqrt{6}}{9}$ 3) $\frac{12\sqrt{3}}{5}$ 4) $\frac{14\sqrt{3}}{5}$

10. **If $ax^2+by^2+2gx+2fy+c=0$ represents an ellipse, then:**

1) Its major axis is parallel to x-axis
2) Its major axis is parallel to y-axis
3) Its axes (i.e. major axis and minor axis)are neither parallel to x-axis nor parallel to y-axis
4) Its axes are parallel to co-ordinate axes

11. **If ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is rotated 90^0 about origin, the equation of ellipse after rotation is:**

1) $\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$ 2) $\frac{x^2}{a(a+b)}+\frac{y^2}{b(a+b)}=1$

3) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ 4) $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

TANGENTS & NORMALS

12. **Tangents to the ellipse $b^2x^2+a^2y^2=a^2b^2$ makes angles θ_1 and θ_2 with major axis such that $\cot\theta_1+\cot\theta_2=k$. Then the locus of the point of intersection is**

1) $xy=2k(y^2+b^2)$ 2) $2xy=k(y^2-b^2)$
3) $4xy=k(y^2-b^2)$ 4) $8xy=k(y^2-b^2)$

13. The locus of the foot of the perpendicular from the centre of the ellipse $x^2+3y^2=3$ on any tangent to it is (JEE-MAINS-2014)

1) $(x^2+y^2)^2=5x^2+7y^2$

2) $(x^2+y^2)^2=7x^2+5y^2$

3) $(x^2+y^2)^2=x^2+3y^2$

4) $(x^2+y^2)^2=3x^2+y^2$

14. Slope of the common tangent to the ellipses $\frac{x^2}{a^2+b^2}+\frac{y^2}{b^2}=1, \frac{x^2}{a^2}+\frac{y^2}{a^2+b^2}=1$ is

1) $\frac{2a}{b}$ 2) $\frac{2b}{a}$ 3) $\frac{a}{b}$ 4) $\frac{b}{a}$

15. If the normal at $p(\theta)$ to $\frac{x^2}{14}+\frac{y^2}{5}=1$ intersect its again $Q(2\theta)$ then $\cos\theta=$

1) $\frac{2}{3}$ 2) $-\frac{2}{3}$ 3) $\frac{3}{2}$ 4) $-\frac{3}{2}$

16. Tangents are drawn to the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$, at ends of latusrecta. The area of quadrilateral so formed is

1) 27 2) $\frac{27}{2}$ 3) $\frac{27}{4}$ 4) $\frac{27}{55}$

17. The points on the ellipse $3x^2+y^2=37$, where the normals to it are perpendicular to $6x+5y-2=0$ are

1) $(3,5);(-3,-5)$ 2) $(2,5);(-2,-5)$

3) $(5,3);(-5,-3)$ 4) $(-5,3);(5,-3)$

LOCUS

18. Equation to the locus of the point which moves such that the sum of its distances from (-4,3) and (4,3) is 12 is

1) $\frac{x^2}{36}+\frac{(y-3)^2}{20}=1$ 2) $\frac{x^2}{20}+\frac{(y-3)^2}{36}=1$

3) $\frac{(x-3)^2}{36}+\frac{y^2}{20}=1$ 4) $\frac{(x-1)^2}{36}+\frac{(y-3)^2}{20}=1$

19. Tangents to the ellipse $b^2x^2+a^2y^2=a^2b^2$ make complimentary angles with the major axis. Then the locus of their point of intersection is

1) $x^2+y^2=a^2-b^2$ 2) $x^2-y^2=a^2+b^2$
3) $x^2-y^2=a^2-b^2$ 4) $x^2+y^2=a^2+b^2$

20. The locus of extremities of the latus-rectum of the family of ellipse $b^2x^2+a^2y^2=a^2b^2$ is $(b\in R)$

1) $x^2-ay=a^3$ 2) $x^2-ay=e^2$

3) $x^2\pm ay=a^2$ 4) $x^2+ay=b^2$

21. The abscissae of the points on the ellipse $9x^2+25y^2-18x-100y-116=0$ lie between

1) 3, -5 2) -4,6
3) -5,7 4) 2,5

22. S and S^1 are the foci of an ellipse whose eccentricity is $\frac{1}{\sqrt{2}}$. B and B^1 are the ends of minor axis then SBS^1B^1 is

1) Parallelogram 2) Rhombus
3) Square 4) Rectangle

23. The arrangement of the ellipses in ascending order of their eccentricities when $\underline{|SBS^1}$ is given, Where $S\ \&\ S^1$ are foci & B is one end of the minor axis

A. $\underline{|SBS^1}=20^0$ B. $\underline{|SBS^1}=60^0$

C. $\underline{|SBS^1}=30^0$ D. $\underline{|SBS^1}=90^0$

1) A,C,B,D 2) D,B,C,A
3) B,D,C,A 4) A,C,D,B

24. If P is a point on the circle $x^2+y^2=9$. The perpendicular PL to the major axis of the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ meets the ellipse at M then $\frac{ML}{PL}=$

1) $\frac{1}{3}$ 2) $\frac{2}{3}$

3) $\frac{1}{2}$ 4) $\frac{3}{2}$

25. If the mid point of chord of the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ is $(0,3)$ and length of the chord is $\frac{4k}{5}$, then k is

1) 2 2) 4 3) 8 4) 10

26. If S and S' are two foci of the ellipse $16x^2+25y^2=400$ and PSQ is a focal chord such that $SP=16$, then $S'Q=$

1. 20 2. $\frac{70}{9}$ 3. $\frac{74}{9}$ 4. $\frac{74}{11}$

27. One focus and the corresponding directrix of an ellipse are (1,2) and $x-y=5$, its eccentricity is 1/2 then centre is

1)(3,0) 2) (0,3) 3) (3,-3) 4) (0,-3)

28. The major axis and minor axis of an ellipse are respectively $x-2y-5=0$ and $2x+y+10=0$, one end of latusrectum is (3,4), then the foci are

1) $(5,0);(-3,-4)$ 2) $(5,0);(-6,-4)$

3) $(5,0);(-11,-8)$ 4) $(5,0);(11,-4)$

29. Number of points on the ellipse $\frac{x^2}{50}+\frac{y^2}{20}=1$ from which pair of perpendicular tangents may be drawn to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ is

1) 0 2) 2 3) 1 4) 4

LEVEL-II (H.W.) - KEY

01) 3	02) 4	03) 3	04) 2	05) 2	06) 1
07) 4	08) 2	09) 2	10) 4	11) 1	12) 2
13) 4	14) 3	15) 2	16) 1	17) 2	18) 1
19) 3	20) 3	21) 2	22) 3	23) 1	24) 2
25) 3	26) 3	27) 2	28) 3	29) 4	

LEVEL-II (H.W.) - HINTS

1. $2ae=\sqrt{36+64}=10$; C(6,8); P(1,-4)

   a = CP = $\sqrt{25+144}=13$

2. Length of sub normal is $|ym|=3$

3. CB < CS

4. $\tan\frac{\theta}{2}=\frac{b}{a}$; $\tan^2\frac{\theta}{2}=\frac{b^2}{a^2}$

   $\frac{1-\cos\theta}{1+\cos\theta}=\frac{b^2}{a^2}$ $\tan\theta=2\sqrt{2}$

5. Here $a^2+2>a^2+1$

   $\Rightarrow a^2+1=(a^2+1)(1-e^2)$

   $\Rightarrow a^2+1=(a^2+2)\frac{5}{6}$ $\Rightarrow 6a^2+6=5a^2+10$

   $\Rightarrow a^2=10-6=4$ $\Rightarrow a=\pm 2$

   Latus rectum $=\frac{2(a^2+1)}{\sqrt{a^2+2}}=\frac{2\times 5}{\sqrt{6}}=\frac{10}{\sqrt{6}}$

6. Let the equation of the ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, It is given that distance between foci is 2h

   $\Rightarrow$ $2ae=2h$ $\Rightarrow$ $ae=2h$ (i)

   Focal distance of one end of minor axis is $a=k$

   $\Rightarrow$ $b^2=a^2-a^2e^2=k^2-h^2$

   So, the equation of the ellipse is $\frac{x^2}{k^2}+\frac{y^2}{k^2-h^2}=1$.

7. Minor axis of E_2 = legnth of latus rectum of E_1

   $=\left(\frac{2b^2}{a}\right)=\frac{18}{5}$

8. Point lies on $x^2+y^2=9$ (director circle)

9. $\frac{2ab}{a^2m^2+b^2}\sqrt{(a^2m^2+b^2-m^2)(1+m^2)}$ (or)

   Solve the line $y=\sqrt{2}x+2$ and ellipse $x^2+4y^2=16$,

   we get $A=(0,2)$ and $B=\left(\frac{-16\sqrt{2}}{9},\frac{-14}{9}\right)$

   length of chord, $AB=\frac{16\sqrt{6}}{9}$

10. There is no *xy* term so we can make perfect squares in *x* and *y* from there it is clear it's axes are parallel to co-ordinate axes, but whether major axis is parallel to x or parallel to y-axis depends on values of coefficients.

11. Lenght of major axis becomes 2b and lenght of minor axis 2a. So equation is $\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$

12. $m_1+m_2=\frac{2x_1y_1}{x_1^2-a^2}; m_1m_2=\frac{y_1^2-b^2}{x_1^2-a^2}$

13. $a^2x^2+b^2y^2=\left(x^2+y^2\right)^2$

14. $\left(a^2+b^2\right)^2m^2+b^2=a^2m^2+a^2+b^2$

15. Normal at $P(\theta)$ is $\frac{\sqrt{14}x}{\cos\theta}-\frac{\sqrt{5}y}{\sin\theta}=9$

This normal passes through

$Q(2\theta)=\left(\sqrt{14}\cos 2\theta,\sqrt{5}\sin 2\theta\right)$

$\Rightarrow \frac{14\cos 2\theta}{\cos\theta}-\frac{5\sin 2\theta}{\sin\theta}=9$

$\Rightarrow (3\cos\theta+2)(6\cos\theta-7)=0$

$\Rightarrow \cos\theta=-\frac{2}{3},\cos\theta\neq\frac{7}{6}$ $\quad\therefore \cos\theta=\frac{-2}{3}$

16. Tangent at $L\left(ae,\frac{b^2}{a}\right)$ is $ex+y=a$ area $=4\left|\frac{c^2}{2ab}\right|$

17. Verification required points lies on ellipse.

18. $SP+S^1P=2a, SS^1=2ae$

19. $m_1m_2=1$

20. $x=\pm ae, y=\pm\frac{b^2}{a}$

$\Rightarrow y^2=a^2\left(1-e^2\right)^2=a^2\left(1-\frac{x^2}{a^2}\right)^2$

21. $-a\le x-h\le a$; $\quad h-a\le x\le h+a$

22. $e=\cos 45^0$

23. $e=\sin\frac{\theta}{2}$, $\underline{|SBS^1}=\theta$

24. Take $M=B$ (end point of minor-axis)

$\Rightarrow L=C$ and $\frac{ML}{PL}=\frac{b}{a}$

25. Equation of the chord whose mid point is $(0,3)$ is $y=3$ intersects the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ at

$\frac{x^2}{16}=1-\frac{9}{25}=\frac{16}{25}\Rightarrow x=\pm\frac{16}{5}$

$\therefore$ Length of the chord $=\frac{32}{5}$

thus $\frac{4k}{5}=\frac{32}{5}\therefore k=8$

26. We know that $\frac{1}{SP}+\frac{1}{SQ}=\frac{2a}{b^2}$ Then for given ellipse $\frac{1}{16}+\frac{1}{SQ}=2\times\frac{5}{16}=\frac{5}{8}\Rightarrow\frac{1}{SQ}=\frac{9}{16}$ Now

$SQ+S'Q=2a=10\Rightarrow S'Q=10-\frac{16}{9}=\frac{74}{9}$

27. $CS:SZ=e^2:1-e^2$

28. Image of (5,0) with respect to $2x+y+10=0$

29. For ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ equation of director circle is $x^2+y^2=25$. This director circle will cut the ellipse $\frac{x^2}{50}+\frac{y^2}{50}=1$ at 4 points

Hence number of points = 4

LEVEL-III

1. The radius of the circle passing through the points of intersection of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $x^2-y^2=0$ is

1) $\frac{ab}{\sqrt{a^2+b^2}}$ 2) $\frac{\sqrt{2}ab}{\sqrt{a^2+b^2}}$

3) $\frac{a^2-b^2}{\sqrt{a^2+b^2}}$ 4) $\frac{a^2+b^2}{\sqrt{a^2-b^2}}$

2. $\frac{x^2}{r^2-r-6}+\frac{y^2}{r^2-6r+5}=1$ will represents the ellipse, if r lies in the interval

1) $(-\infty,-2)\cup(5,\infty)$ 2) $(3,\infty)$

3) $(-2,3)$ 4) $(1,\infty)$

3. A ray of light along the line $x=3$ is reflected at the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$. The slope of the reflected ray is

1) $\frac{4}{15}$ 2) $\frac{8}{15}$ 3) $\frac{15}{8}$ 4) $\frac{15}{4}$

4. The maximum area of the triangle inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

1) ab 2) $\sqrt{3ab}$ 3) $\frac{3\sqrt{3}}{4}ab$ 4) $\frac{ab}{4}$

5. A point on the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ at a distance equal to the mean of the lengths of the semi major axis and semi minor axis from the centre is

1) $\left(\pm\frac{2\sqrt{91}}{7},\pm\frac{3\sqrt{105}}{14}\right)$ 2) $\left(\pm\frac{2\sqrt{91}}{7},\pm\frac{3\sqrt{105}}{7}\right)$

3) $\left(\pm\frac{2\sqrt{105}}{7},\pm\frac{3\sqrt{91}}{14}\right)$ 4) $\left(\pm\frac{2\sqrt{105}}{14},\pm\frac{3\sqrt{91}}{14}\right)$

6. A ray eminating from the point $(-3,0)$ is incident on the ellipse $16x^2+25y^2=400$ at the point P with ordinate 4. Then the equation of the reflected ray is

1) $4x+3y+12=0$ 2) $4x+3y=12$

3) $3x+4y=16$ 4) $3x+4y+9=0$

7. If $4x^2+y^2=1$, then the maximum value of $12x^2-3y^2+16xy$ is

1) 1 2) 3 3) 5 4) 7

8. If PQ is a focal chord of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ Which passes through $S=(3,0)$ and PS = 2 then length of the chord PQ is equal to

1) 10 2) 5 3) 4 4) 2

9. Equation of the largest circle with centre $(1,0)$ that can be inscribed in the ellipse $x^2+4y^2=16$ is

1) $2x^2+2y^2-4x+7=0$

2) $x^2+y^2-2x+5=0$

3) $3x^2+3y^2-6x-8=0$ 4) $x^2+y^2-2x=0$

10. If p is the length of the perpendicular from a focus upon the tangent at any point P of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and r is the distance of P from the focus , then $\frac{2a}{r}-\frac{b^2}{p^2}$ is equal to

1) 1 2) 2 3) 3 4) 0

11. Given an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$ with foci at S and S' and vertices at A and A'. A tangent is drawn at any point P on the ellipse and let R,R',B,B' respectively be the feet of the perpendiculars drawn from $S.S',A.A'$ on the tangent at P. Then the ratio of the areas of the quadrilaterals $S'R'RS$ and $A'B'BA$ is

1) $e:2$ 2) $e:3$

3) $e:1$ 4) $\sqrt{a^2-b^2}:\sqrt{a}$

12. If P is any point lying on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ whose foci are S and S'. Let $\angle PSS'=\alpha$ and $\angle PS'S=\beta$, then which of the following is not true :

1. $PS+PS'=2a,\ if\ a>b$
2. $PS+PS'=2b,\ if\ a<b$
3. $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{1-e}{1+e}$
4. $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{\sqrt{a^2-b^2}}{b^2}\left(a-\sqrt{a^2-b^2}\right)$

13. A series of concentric ellipses $E_1, E_2, \ldots, E_n$ are drawn such that E_n touches the extremities of the major axis of E_{n-1} and the foci of E_n coincide with the extremities of minor axis of E_{n-1}. If the eccentricity of the ellipses is independent of n, then the value of the eccentricity, is

1) $\frac{\sqrt{5}}{3}$ 2) $\frac{\sqrt{5}-1}{2}$

3) $\frac{\sqrt{5}+1}{2}$ 4) $\frac{1}{\sqrt{5}}$

14. If a quadrilateral formed by four tangents to the ellipse $3x^2+4y^2=12$ is a square then

1) The vertices of the square lie on $y=x$
2) The vertices of the square lie on $x^2+y^2=7$
3) The area of such square is 7π
4) Only two such squares are possible

15. The set of positive values of $m(m\neq 0)$ for which a line with slope 'm' is a common tangent to an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and a parabola $y^2=4ax$ is

1) $(0,1)$ 2) $(3,5)$ 3) $(2,1)$ 4) $(2,3)$

16. Let $P(x_1,y_1)$ and $Q(x_2,y_2), y_1<0, y_2<0$ be the ends of the latusrectum of the ellipse $x^2+4y^2=4$. The equations of parabolas with latusrectum PQ are

1) $x^2+2\sqrt{3}y=3+\sqrt{3}$
2) $x^2-2\sqrt{3}y=3+\sqrt{3}$
3) $x^2+2\sqrt{3}y=3-\sqrt{3}$
4) $x^2-2\sqrt{3}y=3-\sqrt{3}$

17. Point P represents the complex number $z=x+iy$ and point Q complex number $z+\frac{1}{z}$. If P moves on the circle $|z|=2$, then Q moves on the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=k$, then k=

1) $\frac{1}{2}$ 2) $\frac{1}{4}$ 3) 1 4) 4

18. Let P_i and P_i' be the feet of the perpendiculars drawn from foci S, S' on a tangent T_i to an ellipse whose length of semi-major axis is 20, if $\sum_{i=1}^{10}(SP_i)(S'P_i')=2560$, then the value of eccentricity is

1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{3}{5}$ 4) $\frac{4}{5}$

19. An ellipse is described by using an endless staring which is passed over two pins. If the axes are 6 cm and 4 cm the necessary length of the string and the distance between the points respectively in cms is

1) $6, 2\sqrt{5}$ 2) $6, \sqrt{5}$ 3) $4, 2\sqrt{5}$ 4) $4, \sqrt{5}$

20. An ellipse has the point (1, –1) and (2, –1) as its foci and x + y = 5 as one of its tangent then the coordinates of the point where this line touches the ellipse are

1) $\left(\frac{34}{9},\frac{11}{9}\right)$ 2) $\left(\frac{32}{9},\frac{13}{9}\right)$
3) $\left(-\frac{34}{9},\frac{79}{9}\right)$ 4) $\left(-\frac{32}{9},\frac{77}{9}\right)$

21. For all real values of $p\in(-1,1)$, the line $2px+y\sqrt{1-p^2}=1$ touches a fixed ellipse. the eccentricity of the ellipse.

1) $\frac{\sqrt{3}}{2}$ 2) 2 3) 3 4) 4

22. The least distance between a point on $x^2+2y^2=6$ and $x+y-7=0$ is $\frac{k}{\sqrt{2}}$ then $k=$

1) 1 2) 4 3) 3 4) 2

23. Coordinates of the vertices B and C of a triangle ABC are $(2,0)$ and $(8,0)$ respectively. The vertex A is varying in such a way that $4\tan\frac{B}{2}\tan\frac{C}{2}=1$. then locus of A is $\frac{(x-5)^2}{25}+\frac{y^2}{k^2}=1$, find k

1) 1 2) 2 3) 3 4) 4

24. If d_1 is the distance between the foci of $\frac{x^2}{a^2+p_1^2}+\frac{y^2}{b^2+p_1^2}=1$ and d_2 in the distance between the foci of $\frac{x^2}{a^2+p_2^2}+\frac{y^2}{b^2+p_2^2}=1$ then $\frac{d_1}{d_2}=$

1) 1 2) $\frac{p_1^2}{p_2^2}$ 3) $\frac{p_2}{p_1}$ 4) $\frac{p_1}{p_2}$

25. If a variable tangent to the circle $x^2+y^2=1$ intersects the ellipse $x^2+2y^2=4$ at points P and Q, then the locus of the point of intersection of tangents to the ellipse at P and Q is a conic whose

1) eccentricity is $\frac{\sqrt{3}}{2}$ 2) eccentricity is $\frac{\sqrt{5}}{2}$

3) latus-rectum is of length 2 units

4) foci are $\left(\pm 2\sqrt{5},0\right)$

26. If $(\alpha,\alpha-1)$ lies inside the ellipse $16x^2+9y^2-16x=0$ then α lies in the interval $\left(\frac{9}{25},k\right)$. Find k.

1)1 2) 2 3)3 4)4

27. $x-2y+4=0$ is a common tangent to $y^2=4ax$ & $\frac{x^2}{4}+\frac{y^2}{b^2}=1$. Then the value of b and the other common tangent are given by:

1) $b=\sqrt{3}$ 2) $x+2y+4=0$

3) $b=3$ 4) $x-2y-4=0$

28. If the variable line $y=kx+2h$ is tangent to an ellipse $2x^2+3y^2=6$, then locus of $P(h,k)$ is a conic C whose eccentricity is e then the value of $3e^2$ is

1) 7 2) $\sqrt{7}$ 3)1 4)2

29. If a tangent of slope 2 of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is normal to the circle $x^2+y^2+4x+1=0$, then the maximum value of ab is

1) 16 2) 8 3)4 4) $\sqrt{5}$

30. The eccentricity of the ellipse which meets the straight line $\frac{x}{7}+\frac{y}{2}=1$ on the axis of X and the straight line $\frac{x}{3}-\frac{y}{5}=1$ on the axis of Y and whose axis lie along the coordinate axes is

1) $\frac{3\sqrt{2}}{7}$ 2) $\frac{2\sqrt{3}}{7}$ 3) $\frac{\sqrt{3}}{7}$ 4) $\frac{2\sqrt{6}}{7}$

31. A tangent to the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ at any point P meet the line x = 0 at a point Q. Let R be the image of Q in the line y = x, then circle whose extremities of a diameter are Q and R passes through a fixed point. The fixed point is

1) (3, 0) 2) (5, 0) 3) (0, 0) 4) (4, 0)

32. Consider the ellipse

$$\frac{x^2}{f\left(k^2+2k+5\right)}+\frac{y^2}{f\left(k+11\right)}=1$$ **and $f(x)$ is a positive decreasing function, then**

1) The set of values of k, for which the major axis is x-axis is (-3, 2)

2) The set of values of k, for which the major axis is y-axis is $(-\infty,2)$

3) The set of values of k, for which the major axis is y-axis is $(-\infty,-3)\cup(2,\infty)$

4) The set of values of k, for which the major axis is y-axis is $(-3,\infty)$

33. The line $x=t^2$ meets the ellipse $x^2+\frac{y^2}{9}=1$ in the real and distinct points if and only if

1) $|t|<2$ 2) $|t|<1$ 3) $|t|>1$ 4) $|t|>2$

34. If the ellipse $\frac{x^2}{4}+y^2=1$ meets the ellipse $x^2+\frac{y^2}{a^2}=1$ in four distinct points and $a=b^2-5b+7$, then b does not lie in

1) $[4,5]$ 2) $(-\infty,2)\cup(3,\infty)$

3) $(-\infty,0)$ 4) $[2,3]$

35. If p, p' denote the lenghts of the perpendiculars from the focus and the centre of an ellipse with semi major axis of lenght a respectively on a tangent to the ellipse and r denote the focal distance of the point, then

1) $rp=ap'$ 2) $ap'=rp+1$

3) $ap=rp'$ 4) $ap=rp'-1$

LEVEL - III - KEY

01) 2	02) 1	03) 2	04) 3	05) 1	06) 2
07) 3	08) 1	09) 3	10) 1	11) 3	12) 4
13) 2	14) 2	15) 1	16) 2,3	17) 2	18) 3
19) 1	20) 1	21) 1	22) 2	23) 4	24) 1
25) 1,3	26) 1	27) 1,2	28) 1	29) 3	30) 4
31) 3	32) 1,3	33) 2	34) 4	35) 3	

LEVEL - III HINTS

1. Two curves are symmetrical about both axes and intersect in four points, so , the circle through their points of intersection will have centre at origin.

Solving $x^2 - y^2 = 0$ and $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$,

we get $x^2 = y^2 = \dfrac{a^2b^2}{a^2+b^2}$

$\therefore$ radius of circle $= \sqrt{\dfrac{2a^2b^2}{a^2+b^2}} = \dfrac{\sqrt{2}ab}{\sqrt{a^2+b^2}}$

2. $r^2 - r - 6 > 0$ and $r^2 - 6r + 5 > 0$

$\Rightarrow (r-3)(r+2) > 0$ and $(r-1)(r-5) > 0$

$\Rightarrow (r < -2 \text{ or } r > 3)$ and $(r < 1 \text{ or } r > 5)$

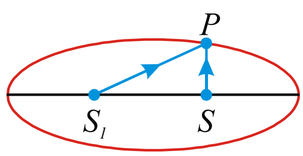
$\Rightarrow r < -2 \text{ or } r > 5$

Also $r^2 - r - 6 \ne r^2 - 6r + 5 \Rightarrow r \ne \dfrac{11}{5}$

3. The foci are $S_1(-3,0), S(3,0)$

The line $x = 3$ intersects the ellipse

$\dfrac{y^2}{16} = 1 - \dfrac{9}{25} = \dfrac{16}{25}, y = \dfrac{16}{5}$



The ray from $S(3,0)$ to $P\left(3, \dfrac{16}{5}\right)$ is reflected at P and passes through $S_1(-3,0)$.

$PS_1 = \dfrac{16}{5} \times \dfrac{1}{(3+3)} = \dfrac{8}{15}$

4. Consider the triangle PQR inscribed in the ellipse with vertices having eccentric angles $\theta_1, \theta_2, \theta_3$ respectively.

$$\Delta PQR = \frac{1}{2}\begin{vmatrix} a\cos\theta_1 & b\sin\theta_1 & 1 \\ a\cos\theta_2 & b\sin\theta_2 & 1 \\ a\cos\theta_3 & b\sin\theta_3 & 1 \end{vmatrix}$$

$$= \frac{ab}{2}\begin{vmatrix} \cos\theta_1 & \sin\theta_1 & 1 \\ \cos\theta_2 & \sin\theta_2 & 1 \\ \cos\theta_3 & \sin\theta_3 & 1 \end{vmatrix} \text{ But } \frac{1}{2}\begin{vmatrix} \cos\theta_1 & \sin\theta_1 & 1 \\ \cos\theta_2 & \sin\theta_2 & 1 \\ \cos\theta_3 & \sin\theta_3 & 1 \end{vmatrix}$$

is the area of triangle inscribed in the circle $x^2 + y^2 = 1$. Its area is maximum if it is equilateral of side $2\sin 60^0 = \sqrt{3}$

The maximum area is $\dfrac{\sqrt{3}}{4}\left(\sqrt{3}\right)^2 = \dfrac{3\sqrt{3}}{4}$

$\therefore$ The maximum area of ΔPQR is $\dfrac{3\sqrt{3}}{4}ab$

5. Lengths of semi-major axis and semi-minor axis of the ellipse $\dfrac{x^2}{16}+\dfrac{y^2}{9}=1$ are 4 and 3 respectively.

So that the mean of these length is $\dfrac{7}{2}$.

Let the coordinates of any point on the ellipse be $P(4\cos\theta, 3\sin\theta)$ .If the distance of P from the centre $O(0,0)$ of the ellipse is $\dfrac{7}{2}$, then

$16\cos^2\theta + 9\sin^2\theta = \dfrac{49}{4}$.

$\Rightarrow 28\cos^2\theta = 13 \Rightarrow \cos\theta = \pm\sqrt{\dfrac{13}{28}} = \pm\dfrac{\sqrt{91}}{14}$

and $\sin\theta = \pm\dfrac{\sqrt{105}}{14}$. So, the coordinates of the required point are $\left(\pm\dfrac{4\sqrt{91}}{14}, \pm\dfrac{3\sqrt{105}}{14}\right)$

i.e. $\left(\pm\dfrac{2\sqrt{91}}{7}, \pm\dfrac{3\sqrt{105}}{14}\right)$

6. $\frac{x^2}{25}+\frac{y^2}{16}=1$; $e=\frac{3}{5}$ foci $(\pm 3,0)$;

$S^{/}(-3,0)$ $S(3,0)$

$P(h,4)$ lies on $\frac{x^2}{25}+\frac{y^2}{16}=1 \Rightarrow P(0,4)$

Required line SP.

7. Put $x=\frac{1}{2}\cos\theta$ $y=\sin\theta$

$12x^2-3y^2+16xy=3\cos 2\theta+4\sin 2\theta\le 5$

8. $\frac{1}{SP}+\frac{1}{SQ}=\frac{2}{l}$ Find SQ, Hence $PQ=SP+SQ$

9. Given ellipse is $\frac{x^2}{16}+\frac{y^2}{4}=1$ ....(i)

Equation of a circle centered at $(1,0)$ can be written as $(x-1)^2+y^2=r^2$ .....(ii)

The abscissae of the intersection points of the circle and the ellipse is given by the equation

$(x-1)^2+\frac{16-x^2}{4}=r^2$

i.e. $4(x^2-2x+1)+16-x^2=4r^2$

i.e. $3x^2-8x+20-4r^2=0$

If the circle lies inside the ellipse, then the roots of the above equation must be imaginary or equal i.e.

$D\le 0$ i.e. $64+12(4r^2-20)\le 0 \Rightarrow r=\le\sqrt{\frac{11}{3}}$

Hence, greatest value of $r=\sqrt{\frac{11}{3}}$ and the equation of required circle is $(x-1)^2+y^2=\frac{11}{3}$

i.e. $3(x^2+y^2)-6x-8=0$

10. The equation of the tangent at $P(a\cos\theta, b\sin\theta)$ on the given ellipse is $\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1$. Length of perpendicular from the focus $(ae,0)$ is

$$p=\frac{e\cos\theta-1}{\sqrt{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}}$$

$$=\frac{ab(e\cos\theta-1)}{\sqrt{b^2\cos^2\theta+a^2(1-\cos^2\theta)}}$$

$$=\frac{ab(e\cos\theta-1)}{\sqrt{a^2-a^2e^2\cos^2\theta}}=-b\sqrt{\frac{1-e\cos\theta}{1+e\cos\theta}}$$

$$\Rightarrow \frac{b^2}{p^2}=\frac{1+e\cos\theta}{1-e\cos\theta}$$

Now $r^2=(ae-a\cos\theta)^2+b^2\sin^2\theta$

$$=a^2\left[(a-\cos\theta)^2+(1-e^2)\sin^2\theta\right]$$

$$=a^2\left[e^2\cos^2\theta-2e\cos\theta+1\right]$$

$$=a^2(1-e\cos\theta)^2 \quad \Rightarrow r=a(1-e\cos\theta)$$

Now $\frac{2a}{r}-\frac{b^2}{p^2}=\frac{2}{1-e\cos\theta}-\frac{1+e\cos\theta}{1-e\cos\theta}=1$

11. Veification : Draw a tangent at B Now $S^{/}R^{/}RS=2ae\times b$, $A^{/}B^{/}BA=2ab$ ratio e : 1

12. $SP+S^1P=2a(a>b), SP+S^1P=2b(a<b)$

$\therefore$ (1) is true, (2) is true

2S[perimeter] = SP+S^{1}P+SS^1 =2a+2ae =2a(1+e)

(3) $\Rightarrow \tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\sqrt{\frac{(S-b)(S-c)}{S(S-c)}}\sqrt{\frac{(S-a)(S-c)}{S(S-b)}}$

$$=\frac{S-c}{S}=\frac{2S-2c}{2S}=\frac{2a(1+e)-2c}{2a(1+e)}$$

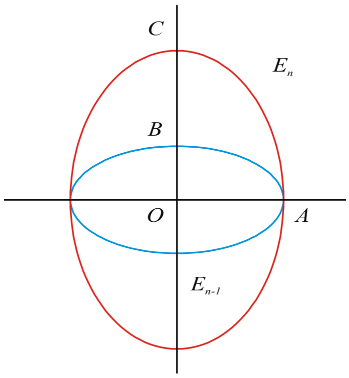
$$=\frac{2a+2ae-4ae}{2a(1+e)}=\frac{2a(1-e)}{2a(1+e)}$$

$\therefore \tan\frac{\alpha}{2}.\tan\frac{\beta}{2}=\frac{1-e}{1+e}$, (4) is false

13. The figure shows two ellipse E_{n-1} and E_n. The Eccentricity is given to be independent of n implies the ratio of minor axis to the major axis, is same for all the ellipse. For ellipse E_{n-1}, let

Minor axis = b, major axis = a

For ellipse E_n, we have



minor axis = a, major axis(OC) = $\frac{OB}{e} = \frac{b}{e}$

[$\because$ B is the focus of E_n]

assuming e to be the eccentricity. Thus, we have

$\frac{b}{a} = \frac{a}{b/e} \Rightarrow e = \frac{b^2}{a^2} = 1 - e^2 \Rightarrow e^2 + e - 1 = 0$

gives $e = \frac{\sqrt{5}-1}{2}$ [$\because$ e must be + ve]

14. Diagnolas of the square are the perpendicular diameters of director circle

15. Tangent to the parabola $y = mx + \frac{a}{m}$

$$c^2 = a^2m^2 + b^2 \Rightarrow \frac{a^2}{m^2} = a^2m^2 + b^2$$

$$\left(\frac{1}{m^2} - m^2\right) = \frac{b^2}{a^2} > 0 \ ; \ 1 - m^4 > 0$$

$$\left(m^2 - 1\right)\left(m^2 + 1\right) < 0 \Rightarrow m^2 - 1 < 0$$

$$\Rightarrow -1 < m < 1 \ , \ m \neq 0$$

16. $\frac{x^2}{4} + \frac{y^2}{1} = 1, a = 2, b = 1$

The ends of latusrectum PQ are $\left(\pm\sqrt{a^2 - b^2}, -\frac{b^2}{a}\right)$

$$P = \left(-\sqrt{3}, -\frac{1}{2}\right), Q\left(\sqrt{3}, -\frac{1}{2}\right)$$

The mid point of PQ is $M\left(0, -\frac{1}{2}\right)$

which $P \overset{\sqrt{3}}{\text{———}} M\left(0,-\frac{1}{2}\right) \overset{\sqrt{3}}{\text{———}} Q$ is the focus of the parabola.

The vertex of the parabola is $V\left(0, -\frac{1}{2} \pm \frac{\sqrt{3}}{2}\right)$

$\therefore$ The parabolas are

$$x^2 = \mp 2\sqrt{3}\left(y - \left(-\frac{1}{2} \pm \frac{\sqrt{3}}{2}\right)\right)$$

i.e., $x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$ and

$x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$

17. Let $Q \equiv \alpha + i\beta$. Given $|z| = 2 \Rightarrow x^2 + y^2 = 4$

Now, $\alpha + i\beta = z + \frac{1}{z} = x + iy + \frac{1}{x + iy}$

$$= x + iy + \frac{x - iy}{x^2 + y^2} = x + iy + \frac{x - iy}{4}$$

$$= x + \frac{x}{4} + i\left(y - \frac{y}{4}\right) \quad = \frac{5x}{4} + \frac{i3y}{4}$$

$\therefore \alpha = \frac{5x}{4}, \beta = \frac{3y}{4}$ Since $x^2 + y^2 = 4$

$\therefore \frac{16\alpha^2}{25} + \frac{16\beta^2}{9} = 4$ or $\frac{\alpha^2}{25} + \frac{\beta^2}{9} = \frac{1}{4}$

Hence Q moves on the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = \frac{1}{4}$

18. $\sum_{i=1}^{10}(SP_i)(S'P'_i) = 2560$

$$\Rightarrow 10b^2 = 2560$$

$$\Rightarrow b^2 = 256 \Rightarrow b = 16 \Rightarrow 256 = 400\left(1 - e^2\right)$$

$$\Rightarrow 1 - e^2 = \frac{16}{25} \Rightarrow e = \frac{3}{5}$$

19. $2a = 6,\ 2b = 4,\ e = \frac{\sqrt{5}}{3}$ Length of string

$SP + S'P = 2a = 6$

distance between the pins $= SS' = 2ae = 2\sqrt{5}$

20. Image of (1, -1) on $x+y=5$ is Q(6,4)

equation of line $S^1\left(2,1\right)Q\left(6,4\right)$ is

$5x-4y=14$ .... (1) Tangent $x+y=5$ ...... (2)

Solve (1) and (2) $\left(\frac{34}{9},\frac{11}{9}\right)$

21. We know that any line of the form $y=mx\pm\sqrt{a^2m^2+b^2}$ touches an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

The given line is $2px+y\sqrt{1-p^2}=1$

$$\Rightarrow y=\left(\frac{-2p}{\sqrt{1-p^2}}\right)x+\frac{1}{\sqrt{1-p^2}}$$

Comparing this equation with

$y=mx\pm\sqrt{a^2m^2+b^2}$, we get

$m=-\frac{2p}{\sqrt{1-p^2}}$ and $a^2m^2+b^2=\frac{1}{1-p^2}$

On eliminating *m*, we get

for all real values of

for all real values of

for all real

values of

and

and

So, the given line touches the ellipse

or, for all real values of p.

Let e be the eccentricity of the ellipse. Then,

22. Any point on ellipse

23.

Thus, sum of distance of variable point A from two given fixed points B and C is always 10.

therefore, equation of locus of A is

.

24.

25. Let

Equation of chord of contact of w.r.t

ellipse is

it touches

26.

27. is a tangent to

The other tangent must be reflection of in the x-axis since both curves are symmetrical about x-axis.

is the other common tangent.

28. By using condition of tangency, we get

Locus of is (which is hyperbola)

Hence

29. Equation of tangent is

this is normal to the circle

this tangent passes through

Using A.M. G.M, we get

30. $a=7$, $b=5$ $e=\sqrt{1-\frac{25}{49}}=\frac{\sqrt{24}}{7}$

31. Equation of the tangent to the ellipse at P $(5\cos\theta,$ $4\sin\theta)$ is $\frac{x\cos\theta}{5}+\frac{y\sin\theta}{4}=1$

It meets the line $x=0$ at Q $(0, 4\operatorname{cosec}\theta)$

Image of Q in the line $y=x$ is R $(4\operatorname{cosec}\theta, 0)$

$\therefore$ Equation of the circle is

$x(x-4\operatorname{cosec}\theta)+y(y-4\operatorname{cosec}\theta)=0$

i.e. $x^2+y^2-4(x+y)\operatorname{cosec}\theta=0$

$\therefore$ each member of the family passes through the intersection of $x^2+y^2=0$ and $x+y=0$

i.e. the point (0, 0).

32. $f(x)$ is a decreasing function and for major axis to be x-axis, $f\left(k^2+2k+5\right)>f\left(k+11\right)$

$\Rightarrow k^2+2k+5<k+11 \Rightarrow k\in(-3,2)$

Then for the remaining values of k, i.e., $k\in(-\infty,-3)\cup(2,\infty)$, major axis is y-axis

33. Solving the given line and the ellipse, we get

$t^2+\frac{y^2}{9}=1 \quad \Rightarrow y^2=9\left(1-t^2\right),$

which gives real and distinct values of y, if $1-t^2>0$

$\Rightarrow t\in(-1,1)$ i.e., $|t|<1$

34. For the two ellipses to intersect at four distinct points, a>1

does not lie in [2, 3]

35. Tangent to the ellipse at

is

= distance from focus

$$\therefore p=\frac{1-e\cos\theta}{\sqrt{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}}$$

Also, p^1 = Distance from centre

$$\Rightarrow p^1=\frac{1}{\sqrt{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}}$$

Again, $r=SP=a(1-e\cos\theta)$

$$\therefore ap=\frac{a-ae\cos\theta}{\sqrt{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}}=rp^1$$

LEVEL - IV

ASSERTION AND REASON-TYPE

Each question contains STATEMENT - 1 (Assertion)and STATEMENT -2 (Reason). Each question has 4 choice (A), (B), (C) and (D) out of which ONLY ONE is correct

Instructions:

1) Statement -1 is True, Statement -2, is True; Statement -2 is a correct explanation for Statement -1

2) Statement -1 is True, Statement -2 is True; Statement -2 Not a correct explanation for Statement -1

3) Statement -1 is True, Statement -2 is False

4) Statement -1 is False, Statement -2 is True.

1. **Statement 1 :** In a triangle ABC , if base BC is fixed and perimeter of the triangle is constant , then vertex A moves on an ellipse
Statement 2 : If the sum of distances of a point P from two fixed points is constant , then locus of P is a real ellipse.

2. **Statement 1:** The equation of the tangents drawn at the ends of the major axis of the ellipse are
Statement 2 : The equation of the tangent drawn at the ends of major axis of the ellipse always parallel to y-axis.

3. **Statement 1:** There can be maximum two points on the line from which perpendicular tangents can be drawn to the ellipse
Statement 2 : Circle and the given line can intersect in maximum two distinct points.

4. **Statement 1 :** The area of the ellipse $2x^2+3y^2=6$ is more than the area of the cicle $x^2+y^2-2x+4y+4=0$.
Statement 2 : The length of semi-major axis of an ellipse is more than the radius of the circle.

5. **Statemen 1 :** If line $x+y=3$ is a tangent to an ellipse with foci $(4,3)$ and $(6,y)$ at the point $(1,2)$ then $y=17$.
Statement 2 : Tangent and normal to the ellipse at any point bisects angle subtended by foci at that point.

6. **Statement 1:** Diagonals of any parallelogram incribed in an ellipse always intersect at the centre is the centre of the ellipse.
Statement 2: Centre of the ellipse is the point at which chord passing through the centre of the ellipse gets bisected at the centre.

7. **Statement - 1 :** From the point $(\lambda,3)$ tangents are drawn to $\frac{x^2}{9}+\frac{y^2}{4}=1$ and are perpendicular to each other then $\lambda=\pm 2$
Statement - 2 : The locus of point of intersection of perpendicular tangents to $\frac{x^2}{9}+\frac{y^2}{4}=1$ is $x^2+3y=13$

8. **Statement-I :** An equation of a common tangent to the parabola $y^2=16\sqrt{3}x$ and the ellipse $2x^2+y^2=4$ is $y=2x+2\sqrt{3}$.
Statement-II : If the line , is a common tangent to the parabola $y^2=16\sqrt{3}x$ and the ellipse $2x^2+y^2=4$, then m satisfies $m^4+2m^2=24$. (2012)

9. **Match the following loci for the ellipse**

COLUMN - I
A) Locus of point of intersection of two perpendicular tangents
B) Locus of foot of perpendicular from any focus upon any tangent
C) Locus of foot of the perpendicular from centre on any tangent
D) Locus of mid point of segment OM where M is the foot of the perpendicular from centre O to any tangent?

COLUMN - II
p)
q)
r)
s)
1)
2)
3)
4)

10. **Match the following**

The correct match is

1)

2)

3)

4)

LEVEL - IV - KEY

1) 3	2)3	3)1	4)2	5) 1	6)1
7)3	8)1	9)4	10)3		

LEVEL - IV - HINTS

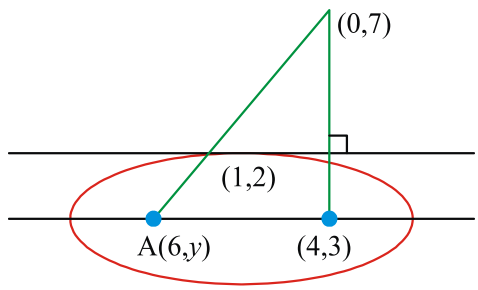
1. Statement 2 is false (locus of P may be a line segment also) . statement 1 is true.

2. , Ends of the major axis are and . Equation of tangent at and is and . Hence, statement 1 is true. But statement 2 is false, as tangents at the ends of major axis may be lines parallel to y-axis when .

3. Locus of point of intersection of perpendicular tangents is director circle,which is . Now line may intersect this circle maximum at two points. Thus there can be maximum two points on the line from which perpendicular tangents can be drawn to the ellipse

4. Given ellipse is , whose area is
Circle is or . Its area is .Hence statement 1 is true .Also statement 2 is true but it is not the correct explanation of statement 1. whose area is and circle whose area is .Also here semi-major axis of ellipse is more than the radius of the circle

5. Statement 2 is true, see theory.
Image of in the line is



and Now points and are collinear.

6. Statement 2 is correct as ellipse is a central conic and it also explains statement

7. Locus of point of intersection of perpendicular tangents to is director circle lie on

8. Statement - I & Statement - II
Verify for parabola and for ellipse

9. A) director circle
B) Auxiliary circle

10. A)Ponits are and Also distance between foci.
B) We known that
C) If the line touches the ellipse ,then
D) Sum of the distance of a point on the ellipse from the foci

* * *